

Technical Information

Proline Promag 10L

Electromagnetic flowmeter



The flowmeter with a weight-optimized sensor and highly cost-effectiveness

Application

- The measuring principle is virtually independent of pressure, density, temperature and viscosity
- Fully suitable for standard applications in the water and wastewater industry

Device properties

- Up to 30 % less sensor weight
- Nominal diameter: DN 25 to 2400 (1 to 90")
- Maximum reduced installation length to DVGW/ISO
- 2-line display with push buttons
- Device in compact or remote version
- HART

Your benefits

- Reduced installation costs – flexible mounting by one-of-a-kind lap-joint flange concept (DN < 350/14")
- Energy-saving flow measurement – no pressure loss due to cross-section constriction
- Maintenance-free – no moving parts
- Cost-effective – designed for easy applications and direct integration
- Safe operation – display provides easy readable process information
- Fully industry compliant – IEC/EN/NAMUR

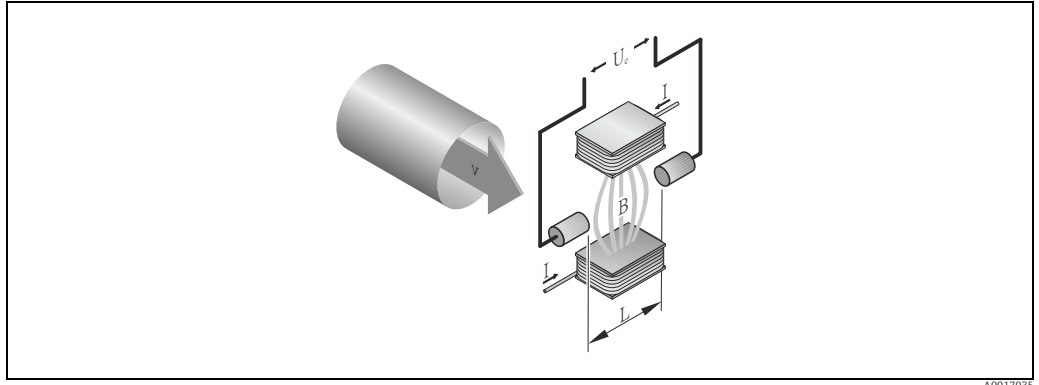
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Function and system design

Measuring principle

Following *Faraday's law of magnetic induction*, a voltage is induced in a conductor moving through a magnetic field.



U_e	Induced voltage
B	Magnetic induction (magnetic field)
L	Electrode spacing
I	Current
v	Flow velocity

In the electromagnetic measuring principle, the flowing medium is the moving conductor. The voltage induced (U_e) is proportional to the flow velocity (v) and is supplied to the amplifier by means of two measuring electrodes. The flow volume (Q) is calculated via the pipe cross-section (A). The DC magnetic field is created through a switched direct current of alternating polarity.

Formulae for calculation:

- Induced voltage $U_e = B \cdot L \cdot v$
- Volume flow $Q = A \cdot v$

Measuring system

The measuring system consists of a transmitter and a sensor.

Two versions are available:

- Compact version: Transmitter and sensor form a mechanical unit.
- Remote version: Sensor is mounted separate from the transmitter.

Transmitter:

- Promag 10 (key operation, two-line, unilluminated display)

Sensor:

- Promag L
 - DN 25 to 300 (1 to 12")
 - DN 350 to 2400 (14 to 90")



Note!

To avoid corrosion, the sensor and process connection material must be selected considering the environmental and process conditions.

Input

Measured variable Flow velocity (proportional to induced voltage)

Measuring range Measuring ranges for liquids
Typically $v = 0.01$ to 10 m/s (0.03 to 33 ft/s) with the specified accuracy
Flow characteristic values (SI units)

Diameter		Recommended flow		Factory settings					
		Min./max. full scale value (v ~ 0.3 or 10 m/s)		Full scale value Current output (v ~ 2.5 m/s)		Pulse value (~ 2 pulses/s)		Low flow cut off (v ~ 0.04 m/s)	
[mm]	[inch]								
25	1"	9 to 300	dm³/min	75	dm³/min	0.50	dm³	1	dm³/min
32	–	15 to 500	dm³/min	125	dm³/min	1.00	dm³	2	dm³/min
40	1 ½"	25 to 700	dm³/min	200	dm³/min	1.50	dm³	3	dm³/min
50	2"	35 to 1100	dm³/min	300	dm³/min	2.50	dm³	5	dm³/min
65	–	60 to 2000	dm³/min	500	dm³/min	5.00	dm³	8	dm³/min
80	3"	90 to 3000	dm³/min	750	dm³/min	5.00	dm³	12	dm³/min
100	4"	145 to 4700	dm³/min	1200	dm³/min	10.00	dm³	20	dm³/min
125	–	220 to 7500	dm³/min	1850	dm³/min	15.00	dm³	30	dm³/min
150	6"	20 to 600	m³/h	150	m³/h	0.025	m³	2.5	m³/h
200	8"	35 to 1100	m³/h	300	m³/h	0.05	m³	5.0	m³/h
250	10"	55 to 1700	m³/h	500	m³/h	0.05	m³	7.5	m³/h
300	12"	80 to 2400	m³/h	750	m³/h	0.10	m³	10	m³/h
350	14"	110 to 3300	m³/h	1000	m³/h	0.10	m³	15	m³/h
375	15"	140 to 4200	m³/h	1200	m³/h	0.15	m³	20	m³/h
400	16"	140 to 4200	m³/h	1200	m³/h	0.15	m³	20	m³/h
450	18"	180 to 5400	m³/h	1500	m³/h	0.25	m³	25	m³/h
500	20"	220 to 6600	m³/h	2000	m³/h	0.25	m³	30	m³/h
600	24"	310 to 9600	m³/h	2500	m³/h	0.30	m³	40	m³/h
700	28"	420 to 13500	m³/h	3500	m³/h	0.50	m³	50	m³/h
750	30"	490 to 15000	m³/h	4000	m³/h	0.50	m³	60	m³/h
800	32"	550 to 18000	m³/h	4500	m³/h	0.75	m³	75	m³/h
900	36"	690 to 22500	m³/h	6000	m³/h	0.75	m³	100	m³/h
1000	40"	850 to 28000	m³/h	7000	m³/h	1.00	m³	125	m³/h
–	42"	950 to 30000	m³/h	8000	m³/h	1.00	m³	125	m³/h
1200	48"	1250 to 40000	m³/h	10000	m³/h	1.50	m³	150	m³/h
–	54"	1550 to 50000	m³/h	13000	m³/h	1.50	m³	200	m³/h
1400	–	1700 to 55000	m³/h	14000	m³/h	2.00	m³	225	m³/h
–	60"	1950 to 60000	m³/h	16000	m³/h	2.00	m³	250	m³/h
1600	–	2200 to 70000	m³/h	18000	m³/h	2.50	m³	300	m³/h
–	66"	2500 to 80000	m³/h	20500	m³/h	2.50	m³	325	m³/h
1800	72"	2850 to 90000	m³/h	23000	m³/h	3.00	m³	350	m³/h
–	78"	3300 to 100000	m³/h	28500	m³/h	3.50	m³	450	m³/h
2000	–	3400 to 110000	m³/h	28500	m³/h	3.50	m³	450	m³/h
–	84"	3700 to 125000	m³/h	31000	m³/h	4.50	m³	500	m³/h
2200	–	4100 to 136000	m³/h	34000	m³/h	4.50	m³	540	m³/h
–	90"	4300 to 143000	m³/h	36000	m³/h	5.00	m³	570	m³/h
2400	–	4800 to 162000	m³/h	40000	m³/h	5.50	m³	650	m³/h

Flow characteristic values (US units)

Diameter		Recommended flow rate Min./max. full scale value (v ~ 0.3 or 10 m/s)	Factory settings		
[inch]	[mm]		Full scale value Current output (v ~ 2.5 m/s)	Pulse value (~ 2 pulses/s)	Low flow cut off (v ~ 0.04 m/s)
1	25	2.5 to 80 gal/min	18 gal/min	0.20 gal	0.25 gal/min
1 ½"	40	7 to 190 gal/min	50 gal/min	0.50 gal	0.75 gal/min
2"	50	10 to 300 gal/min	75 gal/min	0.50 gal	1.25 gal/min
–	65	16 to 500 gal/min	130 gal/min	1 gal	2.0 gal/min
3"	80	24 to 800 gal/min	200 gal/min	2 gal	2.5 gal/min
4"	100	40 to 1250 gal/min	300 gal/min	2 gal	4.0 gal/min
–	125	60 to 1950 gal/min	450 gal/min	5 gal	7.0 gal/min
6"	150	90 to 2650 gal/min	600 gal/min	5 gal	12 gal/min
8"	200	155 to 4850 gal/min	1200 gal/min	10 gal	15 gal/min
10"	250	250 to 7500 gal/min	1500 gal/min	15 gal	30 gal/min
12"	300	350 to 10600 gal/min	2400 gal/min	25 gal	45 gal/min
14"	350	500 to 15000 gal/min	3600 gal/min	30 gal	60 gal/min
15"	375	600 to 19000 gal/min	4800 gal/min	50 gal	60 gal/min
16"	400	600 to 19000 gal/min	4800 gal/min	50 gal	60 gal/min
18"	450	800 to 24000 gal/min	6000 gal/min	50 gal	90 gal/min
20"	500	1000 to 30000 gal/min	7500 gal/min	75 gal	120 gal/min
24"	600	1400 to 44000 gal/min	10500 gal/min	100 gal	180 gal/min
28"	700	1900 to 60000 gal/min	13500 gal/min	125 gal	210 gal/min
30"	750	2150 to 67000 gal/min	16500 gal/min	150 gal	270 gal/min
32"	800	2450 to 80000 gal/min	19500 gal/min	200 gal	300 gal/min
36"	900	3100 to 100000 gal/min	24000 gal/min	225 gal	360 gal/min
40"	1000	3800 to 125000 gal/min	30000 gal/min	250 gal	480 gal/min
42"	–	4200 to 135000 gal/min	33000 gal/min	250 gal	600 gal/min
48"	1200	5500 to 175000 gal/min	42000 gal/min	400 gal	600 gal/min
54"	–	9 to 300 Mgal/day	75 Mgal/day	0.0005 Mgal	1.3 Mgal/min
–	1400	10 to 340 Mgal/day	85 Mgal/day	0.0005 Mgal	1.3 Mgal/min
60"	–	12 to 380 Mgal/day	95 Mgal/day	0.0005 Mgal	1.3 Mgal/min
–	1600	13 to 450 Mgal/day	110 Mgal/day	0.0008 Mgal	1.7 Mgal/min
66"	–	14 to 500 Mgal/day	120 Mgal/day	0.0008 Mgal	2.2 Mgal/min
72"	1800	16 to 570 Mgal/day	140 Mgal/day	0.0008 Mgal	2.6 Mgal/min
78"	–	18 to 650 Mgal/day	175 Mgal/day	0.0010 Mgal	3.0 Mgal/min
–	2000	20 to 700 Mgal/day	175 Mgal/day	0.0010 Mgal	2.9 Mgal/day
84"	–	24 to 800 Mgal/day	190 Mgal/day	0.0011 Mgal	3.2 Mgal/day
–	2200	26 to 870 Mgal/day	210 Mgal/day	0.0012 Mgal	3.4 Mgal/day
90"	–	27 to 910 Mgal/day	220 Mgal/day	0.0013 Mgal	3.6 Mgal/day
–	2400	31 to 1030 Mgal/day	245 Mgal/day	0.0014 Mgal	4.1 Mgal/day

Operable flow range

Over 1000 : 1

Output

Output signal	Current output - Galvanically isolated - Active: 4 to 20 mA, $R_L < 700 \Omega$ (for HART: $R_L \geq 250 \Omega$) - Full scale value adjustable - Temperature coefficient: typ. $2 \mu A/^{\circ}C$, resolution: $1.5 \mu A$ Pulse/status output - Galvanically isolated - Passive: 30 V DC / 250 mA - Open collector - Can be configured as: - Pulse output: Pulse value and pulse polarity can be selected, max. pulse width adjustable (5 to 2000 ms), pulse frequency max. 100 Hz - Status output: for example, can be configured for error messages, empty pipe detection, flow recognition, limit value
Signal on alarm	- Current output → Failsafe mode can be selected (e.g. in accordance with NAMUR Recommendation NE 43) - Pulse output → Failsafe mode can be selected - Status output → "Not conductive" in the event of fault or power supply failure
Load	→ Section "output signal"
Low flow cutoff	Switch-on points for low flow are selectable.
Galvanic isolation	All circuits for inputs, outputs and power supply are galvanically isolated from each other.

Power supply

Terminal assignment

Order code for "Input/Output"	Terminal No.					
	24 (+)	25 (-)	26 (+)	27 (-)	1 (L1/L+)	2 (N/L-)
A	Pulse/status output		HART current output		Power supply	
Functional values	→ 6, Section "Output signal"				→ Section "Supply voltage"	

Supply voltage

- 85 to 250 V AC, 45 to 65 Hz
- 20 to 28 V AC, 45 to 65 Hz
- 11 to 40 V DC

Power consumption

- 11 to 40 V DC: < 6 W (incl. sensor)
- 20 to 28 V AC: < 8 VA (incl. sensor)
- 85 to 250 V AC: < 12 VA (incl. sensor)

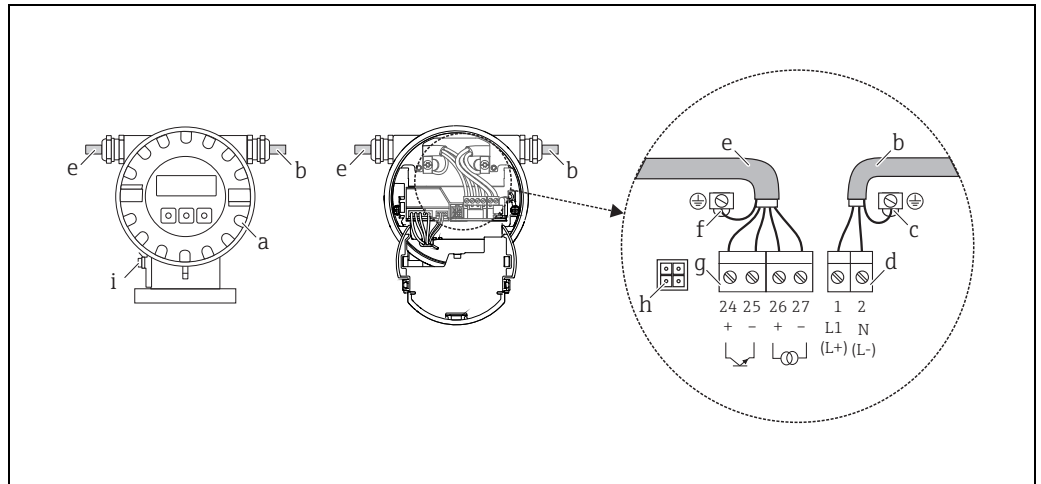
Switch-on current:

- Max. 3.3 A (< 5 ms) for 24 V DC
- Max. 5.5 A (< 5 ms) for 28 V AC
- Max. 16 A (< 5 ms) for 250 V AC

Power supply failure

Lasting min. ½ cycle frequency: EEPROM saves measuring system data

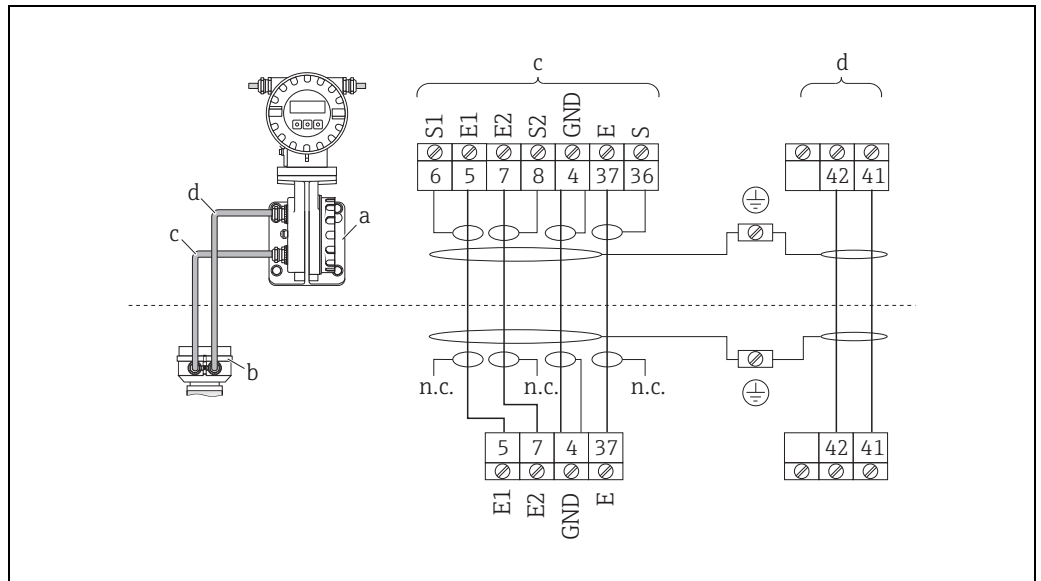
Electrical connection



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Connecting the transmitter (aluminum field housing), cable cross-section max. 2.5 mm² (14 AWG)

- a Electronics compartment cover
- b Power supply cable
- c Ground terminal for power supply cable
- d Terminal connector for power supply cable
- e Electrode cable
- f Ground terminal for electrode cable
- g Terminal connector for electrode cable
- h Service connector
- i Ground terminal for potential equalization

Electrical connection,
remote version

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Connecting the remote version

- a Wall-mount housing connection compartment
- b Sensor connection housing cover
- c Electrode cable
- d Coil current cable
- n.c. Not connected, insulated cable shields

Terminal numbers and cable colors:

5/6 = brown, 7/8 = white, 4 = green, 37/36 = yellow

Potential equalization



Note!

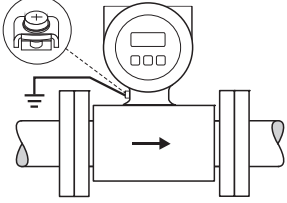
The measuring system must be included in the potential equalization.

Perfect measurement is only ensured when the fluid and the sensor have the same electrical potential. This is ensured by the reference electrode integrated in the sensor as standard.

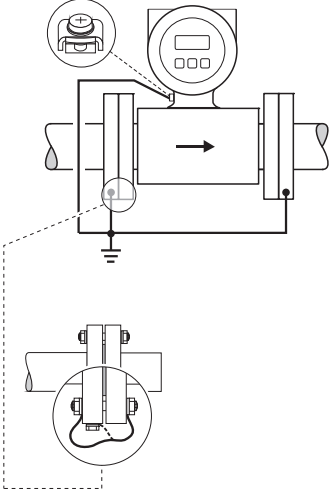
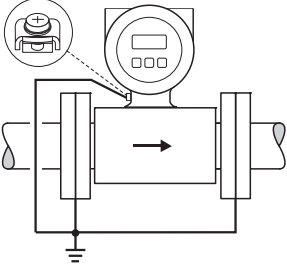
The following should also be taken into consideration for potential equalization:

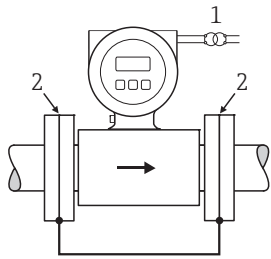
- Internal grounding concepts in the company
- Operating conditions, such as the material/ grounding of the pipes (see table)

Standard situation

Operating conditions	Potential equalization
When using the measuring device in a: ■ Metal, grounded pipe Potential equalization takes place via the ground terminal of the transmitter.  Note! When installing in metal pipes, we recommend you connect the ground terminal of the transmitter housing with the piping.	 A0010831 <i>Via the ground terminal of the transmitter</i>

Special situations

Operating conditions	Potential equalization
When using the measuring device in a: ■ Metal pipe that is not grounded This connection method also applies in situations where: ■ Customary potential equalization cannot be ensured. ■ Excessively high equalizing currents can be expected. Both sensor flanges are connected to the pipe flange by means of a ground cable (copper wire, at least 6 mm² / 0.0093 in²) and grounded. Connect the transmitter or sensor connection housing, as applicable, to ground potential by means of the ground terminal provided for the purpose. The ground cable is mounted directly on the conductive flange coating with the flange screws.  Note! The ground cable for flange-to-flange connections can be ordered separately as an accessory from Endress+Hauser.	 A0011567 <i>Via the ground terminal of the transmitter and the flanges of the pipe</i>
When using the measuring device in a: ■ Plastic pipe ■ Pipe with insulating lining This connection method also applies in situations where: ■ Customary potential equalization cannot be ensured. ■ Excessively high equalizing currents can be expected. Potential equalization takes place using additional ground disks, which are connected to the ground terminal via a ground cable (copper wire, at least 6 mm² / 0.0093 in²). When installing the ground disks, please comply with the enclosed Installation Instructions.	 A0010833 <i>Via the ground terminal of the transmitter and the optionally available ground disks</i>

Operating conditions	Potential equalization
When using the measuring device in a: ■ Pipe with a cathodic protection unit The device is installed potential-free in the pipe. Only the two flanges of the pipe are connected with a ground cable (copper wire, at least 6 mm² / 0.0093 in²). Here, the ground cable is mounted directly on the conductive flange coating with flange screws. Note the following when installing: ■ The applicable regulations regarding potential-free installation must be observed. ■ There should be no electrically conductive connection between the pipe and the device. ■ The mounting material must withstand the applicable torques.	 Potential equalization and cathodic protection 1 Power supply isolation transformer 2 Electrically isolated

Cable entry

Power supply and electrode cables (inputs/ outputs):

- Cable entry M20 × 1.5 (8 to 12 mm / 0.31 to 0.47")
- Thread for cable entries, ½" NPT, G ½"

Connecting cable for remote version:

- Cable entry M20 × 1.5 (8 to 12 mm / 0.31 to 0.47")
- Thread for cable entries, ½" NPT, G ½"

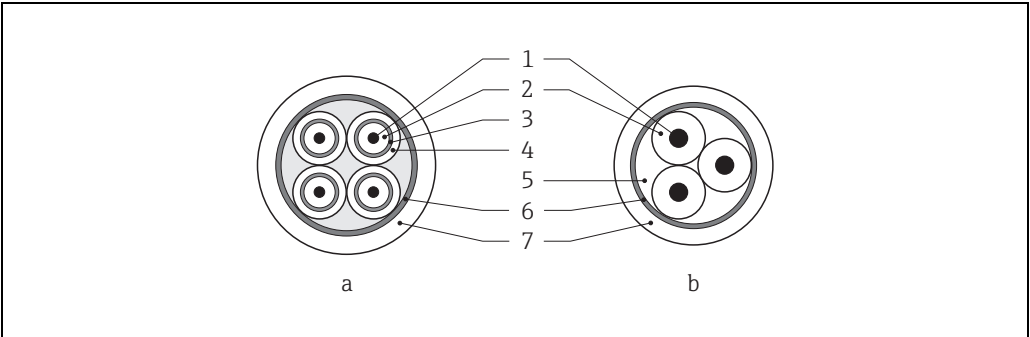
Cable specification

Coil current cable

- 3 × 0.75 mm² (18 AWG) PVC cable with common, braided copper shield (Ø ~ 9 mm / 0.35")
- Conductor resistance: ≤ 37 Ω/km (≤ 0.011 Ω/ft)
- Capacitance core/core, shield grounded: ≤ 120 pF/m (≤ 37 pF/ft)
- Operating temperature: -20 to +80 °C (-68 to +176 °F)
- Cable cross-section: max. 2.5 mm² (14 AWG)
- Test voltage for cable insulation: ≤ 1433 AC r.m.s. 50/60 Hz or ≥ 2026 V DC

Electrode cable

- 3 × 0.38 mm² (20 AWG) PVC cable with common, braided copper shield (Ø ~ 9.5 mm / 0.37") and individual shielded cores
- With empty pipe detection (EPD): 4 × 0.38 mm² (20 AWG) PVC cable with common, braided copper shield (Ø ~ 9.5 mm / 0.37") and individual shielded cores
- Conductor resistance: ≤ 50 Ω/km (≤ 0.015 Ω/ft)
- Capacitance core/shield: ≤ 420 pF/m (≤ 128 pF/ft)
- Operating temperature: -20 to +80 °C (-68 to +176 °F)
- Cable cross-section: max. 2.5 mm² (14 AWG)



- a Electrode cable
b Coil current cable

- 1 Core
2 Core insulation
3 Core shield
4 Core jacket
5 Core reinforcement
6 Cable shield
7 Outer jacket

Operation in zones of severe electrical interference
The measuring device complies with the general safety requirements in accordance with EN 61010 and the EMC requirements of IEC/EN 61326.

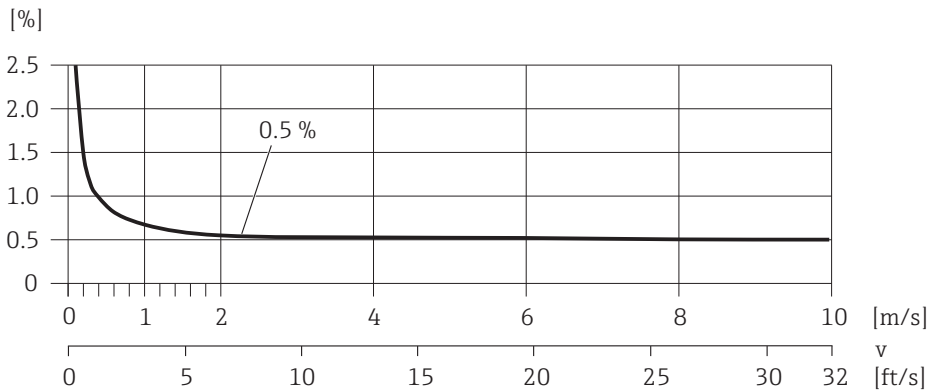


Note!
Grounding is by means of the ground terminals provided for the purpose inside the connection housing. Ensure that the stripped and twisted lengths of cable shield to the ground terminal are as short as possible.

Performance characteristics

- Reference operating conditions
- Error limits following DIN EN 29104, future ISO 20456
 - Water, typically +15 to +45°C (+59 to +113 °F); 0.5 to 7 bar (73 to 101 psi)
 - Specification as per calibration protocol
 - Data on the measured error based on accredited calibration rigs traced back to ISO 17025

- Maximum measured error
- Current output: also typically $\pm 5\ \mu\text{A}$
 - Pulse output: $\pm 0.5\%$ o.r. $\pm 2\ \text{mm/s}$ ($\pm 0.5\%$ o.r. $\pm 0.08\ \text{in/s}$) (o.r. = of reading)
- Fluctuations in the supply voltage do not have any effect within the specified range.



Max. measured error in % of reading

A0003200

- Repeatability
- Max. $\pm 0.2\%$ o.r. $\pm 2\ \text{mm/s}$ ($\pm 0.2\%$ o.r. $\pm 0.08\ \text{in/s}$) (o.r. = of reading)

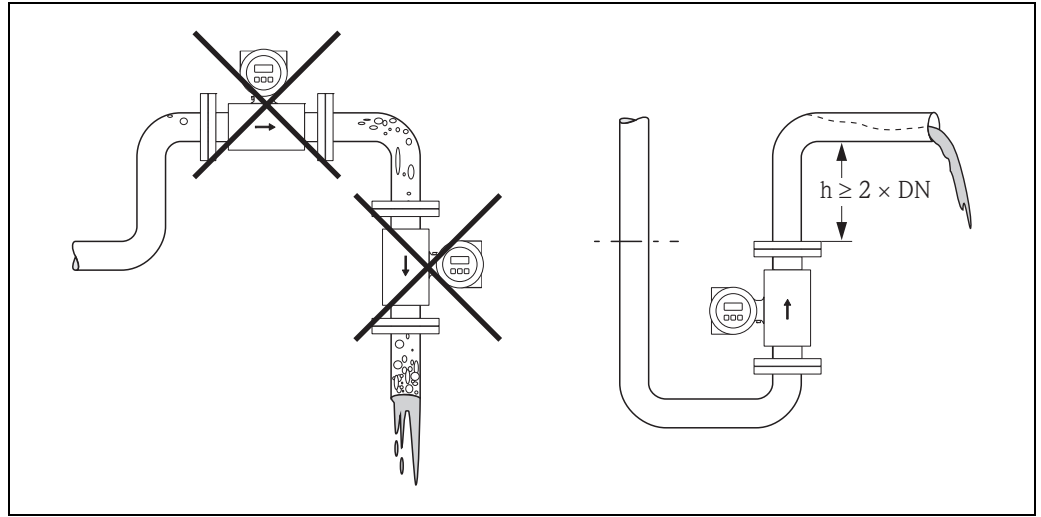
Installation

Mounting location

Entrained air or gas bubble formation in the measuring tube can result in an increase in measuring errors.

Avoid the following installation locations in the pipe:

- Highest point of a pipeline. Risk of air accumulating!
- Directly upstream from a free pipe outlet in a vertical pipeline.



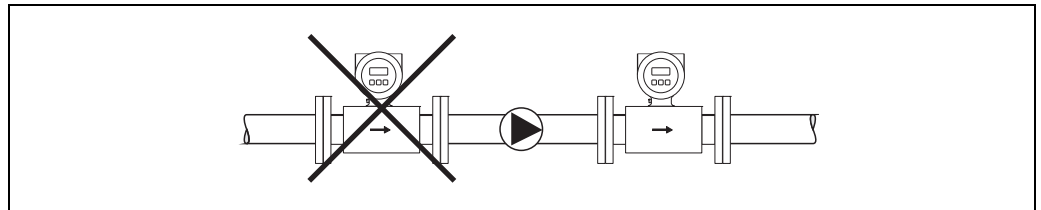
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Mounting location

Installation of pumps

Sensors may not be installed on the pump suction side. This precaution is to avoid low pressure and the consequent risk of damage to the lining of the measuring tube. Information on the pressure tightness of the measuring tube lining → 17, Section "Pressure tightness".

Pulsation dampers may be needed when using piston pumps, piston diaphragm pumps or hose pumps. Information on the shock and vibration resistance of the measuring system → 16, Section "Shock and vibration resistance".



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Installation of pumps

Partially filled pipes

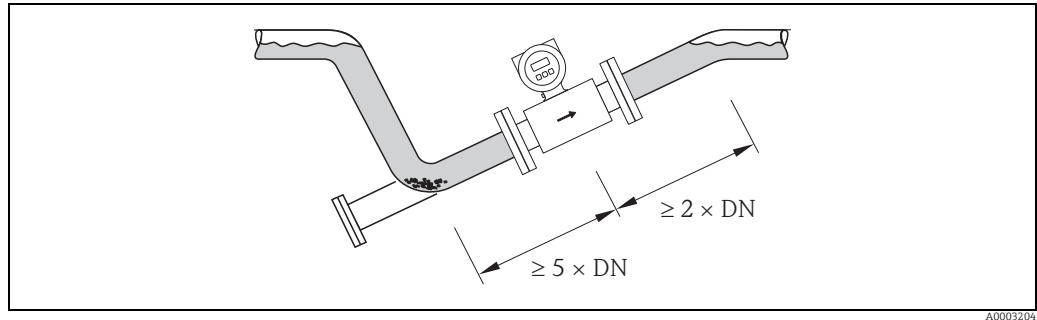
Partially filled pipes with gradients necessitate a drain-type configuration.

The empty pipe detection function (EPD) provides additional security in detecting empty or partially filled pipes.



Note!

Risk of solids accumulating. Do not install the sensor at the lowest point in the drain. It is advisable to install a cleaning valve.

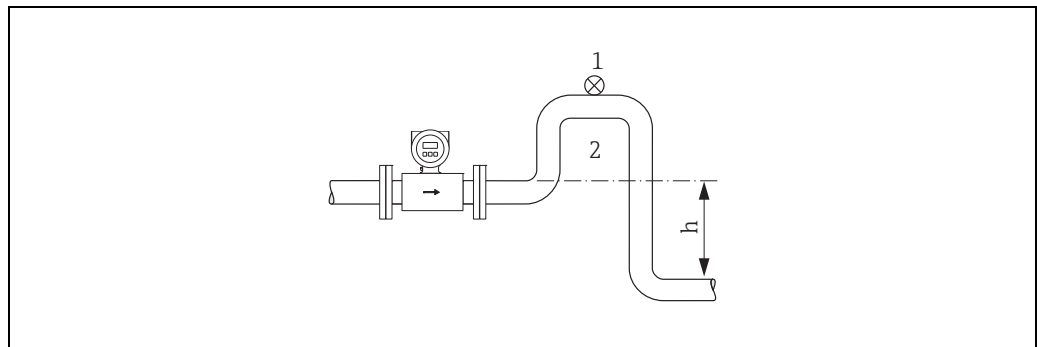


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Installation with partially filled pipes

Down pipes

Install a siphon or a vent valve downstream of the sensor in down pipes $h \geq 5 \text{ m}$ (16.4 ft). This precaution is to avoid low pressure and the consequent risk of damage to the lining of the measuring tube. This measure also prevents the liquid current stopping in the pipe which could cause air locks. Information on the pressure tightness of the measuring tube lining → 17, Section "Pressure tightness".



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Installation measures for vertical pipes

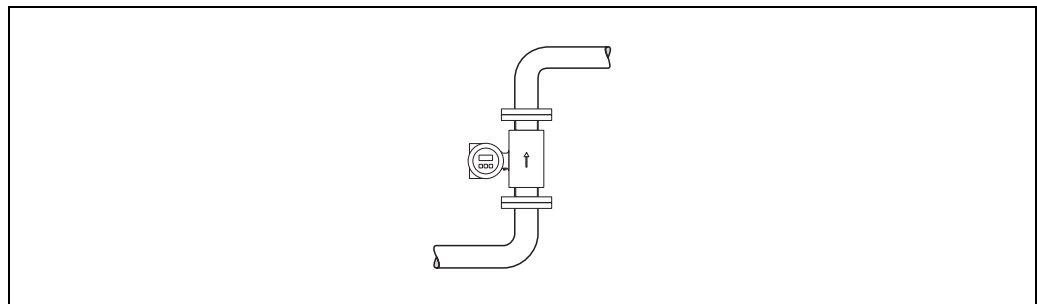
- 1 Vent valve
- 2 Pipe siphon
- h Length of the down pipe

Orientation

An optimum orientation helps avoid gas and air accumulations and deposits in the measuring tube. However, the measuring device also offers the additional function of empty pipe detection (EPD) for detecting partially filled measuring tubes or if outgassing fluids or fluctuating operating pressures are present.

Vertical orientation

This is the ideal orientation for self-emptying piping systems and for use in conjunction with empty pipe detection.



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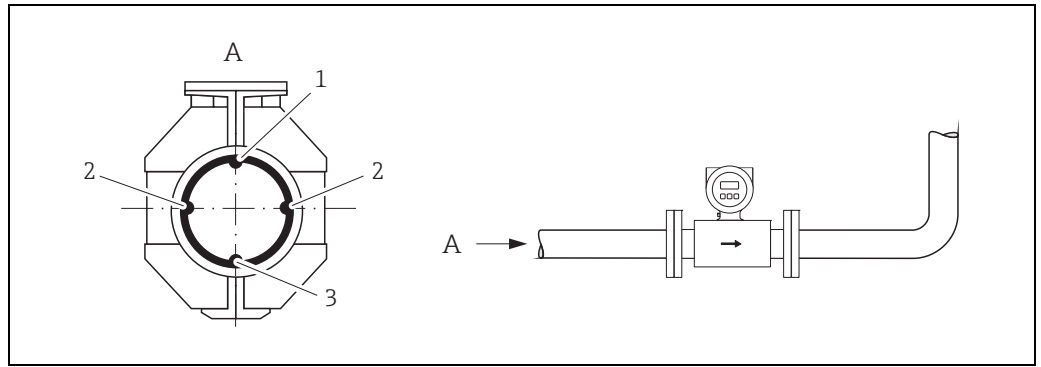
Vertical orientation

Horizontal orientation

The measuring electrode axis should be horizontal. This prevents brief insulation of the two measuring electrodes by entrained air bubbles.

**Note!**

Empty pipe detection only works correctly with horizontal orientation if the transmitter housing is facing upwards. Otherwise there is no guarantee that empty pipe detection will respond if the measuring tube is only partially filled or empty.



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Horizontal orientation

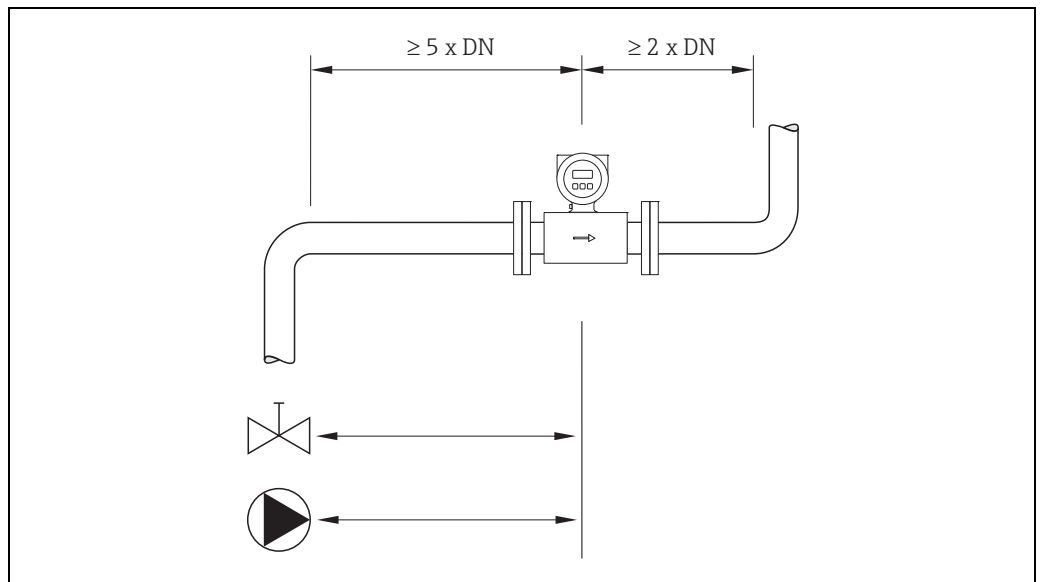
- 1 EPD electrode for empty pipe detection
- 2 Measuring electrodes for signal detection
- 3 Reference electrode for potential equalization

Inlet and outlet run

If possible, install the sensor well clear of assemblies such as valves, T-pieces, elbows etc.

Note the following inlet and outlet runs to comply with measuring accuracy specifications:

- Inlet run: $\geq 5 \times \text{DN}$
- Outlet run: $\geq 2 \times \text{DN}$



A0003210

Inlet and outlet run

Adapters

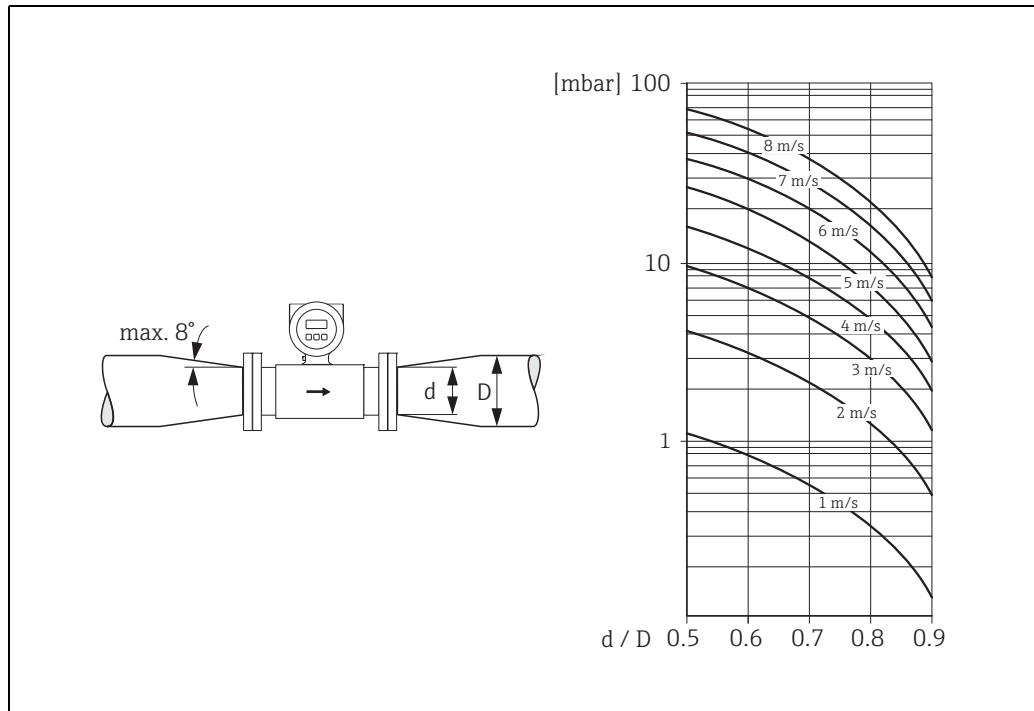
Suitable adapters to DIN EN 545 (double-flange reducers) can be used to install the sensor in larger-diameter pipes. The resultant increase in the rate of flow improves measuring accuracy with very slow-moving fluids. The nomogram shown here can be used to calculate the pressure loss caused by reducers and expanders.



Note!

The nomogram only applies to liquids of viscosity similar to water.

1. Calculate the ratio of the diameters d/D .
2. From the nomogram read off the pressure loss as a function of flow velocity (downstream from the reduction) and the d/D ratio.



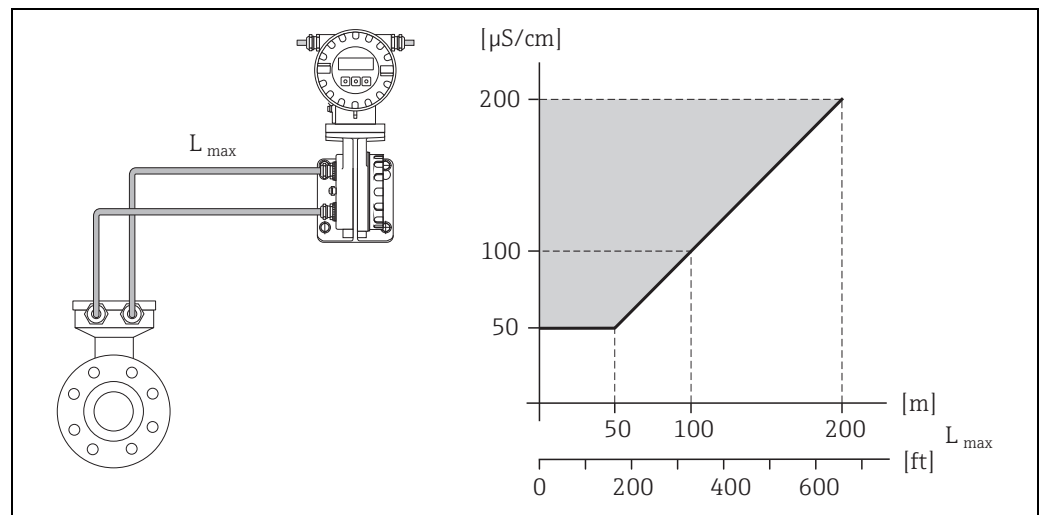
A0003213

Pressure loss due to adapters

Length of connecting cable

When mounting the remote version, please note the following to achieve correct measuring results:

- Fix cable run or lay in armored conduit. Cable movements can falsify the measuring signal especially in the case of low fluid conductivities.
- Route the cable well clear of electrical machines and switching elements.
- If necessary, ensure potential equalization between sensor and transmitter.
- The permitted cable length $L_{\max}$ is determined by the fluid conductivity.
A minimum conductivity of $50 \mu\text{S}/\text{cm}$ is needed for all fluids.
- When the empty pipe detection function is switched on (EPD), the maximum connecting cable length is 10 m (33 ft).



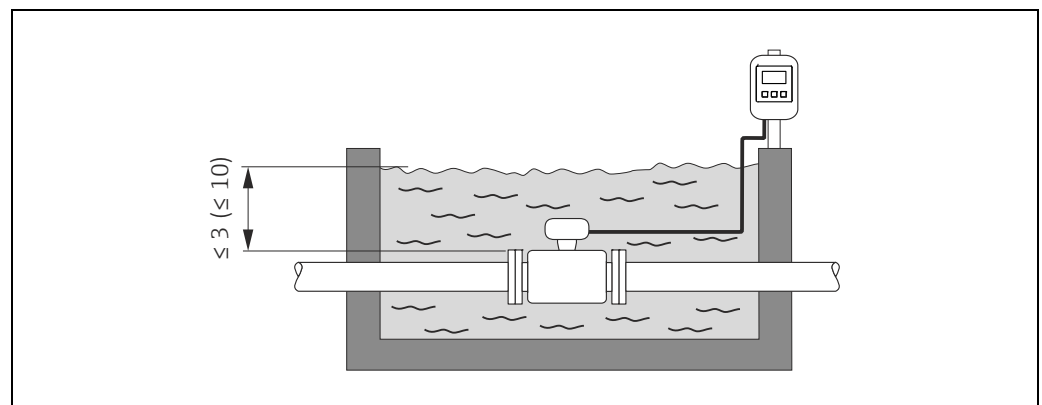
Permitted length of connecting cable for remote version

Area marked in gray = permitted range; $L_{\max}$ = length of connecting cable in [m] ([ft]); fluid conductivity in [$\mu\text{S}/\text{cm}$]

Special mounting instructions**Temporary use in water**

A remote version in IP67, type 6 is optionally available for temporary use in water up to 168 hours at $\leq 3 \text{ m}$ (10 ft) or, in exceptional cases, for use up to 48 hours at $\leq 10 \text{ m}$ (30 ft).

Compared to the degree of protection of standard version IP67, type 4X enclosure, the version IP67, type 6 enclosure was designed to withstand short-term or temporary submergence (e.g. flooding).



Engineering unit m (ft)

Environment

Ambient temperature range

Transmitter

- -20 to +60 °C (-4 to +140 °F)

Sensor

- Flange material carbon steel: -10 to +60 °C (14 to +140 °F)
- Flange material stainless steel (DN ≤ 300/12"): -40 to +60 °C (-40 to +140 °F)



Note!

The permitted temperature range of the measuring tube lining may not be undershot or overshoot (→ 17, Section "Medium temperature range").

Please note the following points:

- Install the device in a shady location. Avoid direct sunlight, particularly in warm climatic regions.
- The transmitter must be mounted separate from the sensor if both the ambient and fluid temperatures are high.

Storage temperature

The storage temperature corresponds to the operating temperature range of the measuring transmitter and the appropriate measuring sensors.



Note!

- The measuring device must be protected against direct sunlight during storage in order to avoid unacceptably high surface temperatures.
- A storage location must be selected where moisture does not collect in the measuring device. This will help prevent fungus and bacteria infestation which can damage the liner.

Degree of protection

Transmitter

- As standard: IP67, type 4X enclosure
- When housing is open: IP20; type 1 enclosure

Sensor

- As standard: IP67, type 4X enclosure
 - Optionally available for remote version:
 - IP67, Type 6P enclosure. Suitable for temporary use in water up to 168 hours at ≤ 3 m (10 ft) or up to 48 hours at ≤ 10 m (30 ft).
 - IP68, Type 6P enclosure (for DN ≤ 300 (12") only possible in conjunction with stainless steel flanges)
- Not suitable for use in corrosive atmospheres/liquids or in buried applications if special precautions are not taken.

Shock and vibration resistance

Acceleration up to 2 g following IEC 600 68-2-6

Electromagnetic compatibility (EMC)

- As per IEC/EN 61326 as well as NAMUR Recommendation NE 21
- Emission: to limit value for industry EN 55011

Process

Medium temperature range

- 0 to +80 °C (+32 to +176 °F) for hard rubber (DN 350 to 2400 / 14 to 90")
- 20 to +50 °C (-4 to +122 °F) for polyurethane (DN 25 to 1200 / 1 to 48")
- 20 to +90 °C (-4 to +194 °F) for PTFE (DN 25 to 300 / 1 to 12")

Conductivity

The minimum conductivity is: $\geq 50 \mu\text{S/cm}$

Note!

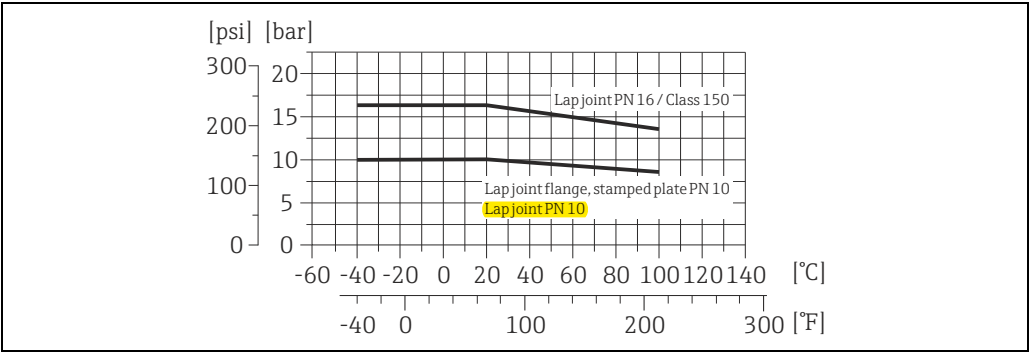
In the remote version, the necessary minimum conductivity also depends on the cable length ($\rightarrow$ 15, Section "Length of connecting cable").

Pressure-temperature ratings

Note!

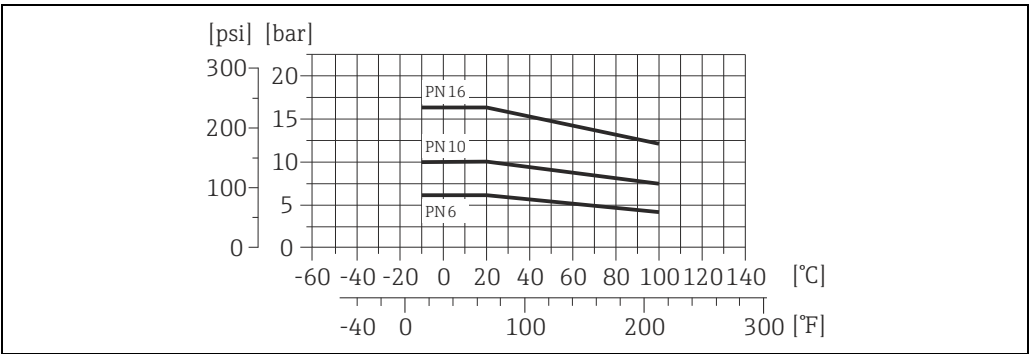
- The following diagrams contain allowable operating pressures as a function of the medium temperature. However, the maximum medium temperatures permitted always depend on the lining material of the sensor and/or the sealing material ($\rightarrow$ 17).
- Permitted test pressure = $1.5 \times$ nominal pressure

Process connection: lap joint flange/lap joint flange, stamped plate according to EN 1092-1 (DIN 2501) and ASME B16.5; DN 25 to 300 (1 to 12")

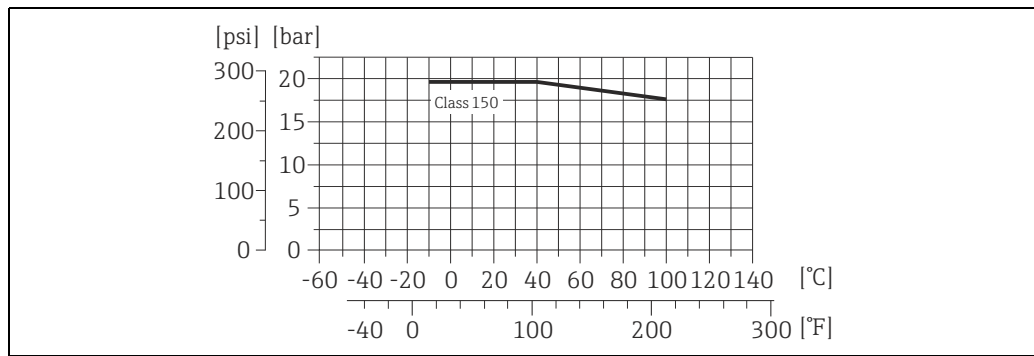


Process connection material: stainless steel (min. -40 °C (-40 °F)), carbon steel (min. -10 °C (+14 °F))

Process connection: flange according to EN 1092-1 (DIN 2501); DN 350 to 2400 (14 to 90")

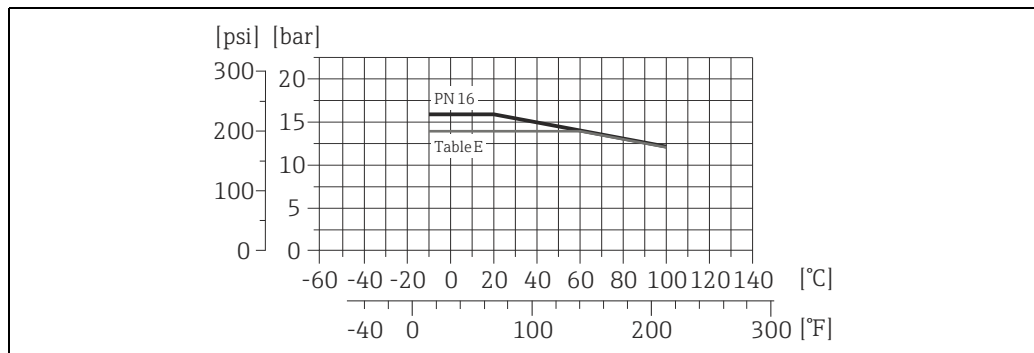


Process connection material: carbon steel

Process connection: flange according to ASME B16.5; DN 350 to 600 (14 to 24")

A0025179-EN

Process connection material: A105

Process connection: flange according to AS 2129 and AS 4087; DN 350 to 2400 (14 to 90")

A0025180-EN

Process connection material: carbon steel

**Medium pressure range
(nominal pressure)**

- EN 1092-1 (DIN 2501)
 - PN 6 (DN 350 to 2400 / 14 to 90")
 - PN 10 (DN 200 to 2400 / 8 to 90")
 - PN 16 (DN 25 to 2000 / 1 to 78")
- EN 1092-1, lap joint flange, stamped plate
 - PN 10 (DN 25 to 300 / 1 to 12")
- ASME B16.5
 - Class 150 (1 to 24")
- AWWA C207
 - Class D (28 to 90")
- AS2129
 - Table E (350 to 1200 / 14 to 48")
- AS4087
 - PN 16 (350 to 1200 / 14 to 48")

Pressure tightness*Measuring tube lining: Polyurethane, hard rubber*

Promag L Nominal diameter		Measuring tube lining	Resistance of measuring tube lining to partial vacuum: Limit values for abs. pressure [mbar] ([psi]) at various fluid temperatures		
[mm]	[inch]		25 °C 77 °F	50 °C 122 °F	80 °C 176 °F
25 to 1200	1 to 48"	Polyurethane	0	0	–
350 to 2400	14 to 90"	Hard rubber	0	0	0

Measuring tube lining: PTFE

Nominal diameter		Limit values for abs. pressure [mbar] ([psi]) at fluid temperatures:			
[mm]	[inch]	25 °C (77 °F)		90 °C (194 °F)	
		[mbar]	[psi]	[mbar]	[psi]
25	1"	0	0	0	0
32	–	0	0	0	0
40	1 ½"	0	0	0	0
50	2"	0	0	0	0
65	–	0	0	40	0.58
80	3"	0	0	40	0.58
100	4"	0	0	135	1.96
125	–	135	1.96	240	3.48
150	6"	135	1.96	240	3.48
200	8"	200	2.90	290	4.21
250	10"	330	4.79	400	5.80
300	12"	400	5.80	500	7.25

Limiting flow

The diameter of the pipe and the flow rate determine the nominal diameter of the sensor. The optimum flow velocity is between 2 to 3 m/s (6.5 to 9.8 ft/s). The velocity of flow (v), moreover, has to be matched to the physical properties of the fluid:

- $v < 2$ m/s (6.5 ft/s): for abrasive fluids such as potter's clay, lime milk, ore slurry etc.
- $v > 2$ m/s (6.5 ft/s): for fluids causing build-up such as wastewater sludges etc.

For an overview of the measuring range full scale values, see the "Measuring range" section →  4.

Pressure loss

- No pressure loss if the sensor is installed in a pipe with the same nominal diameter.
- Pressure losses for configurations incorporating adapters according to DIN EN 545 (→  14, Section "Adapters").

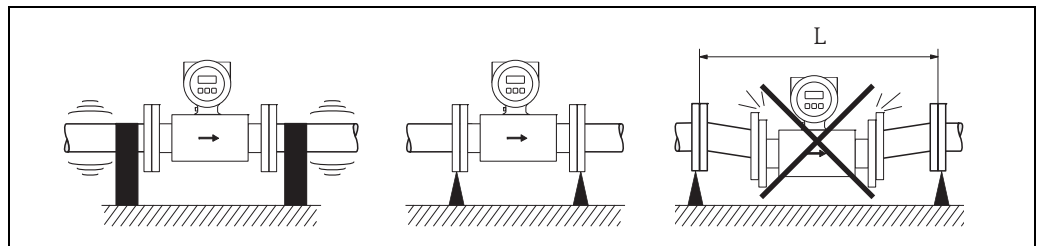
Vibrations



Secure the piping and the sensor if vibration is severe.

Note!

If vibrations are too severe, we recommend the sensor and transmitter be mounted separately. Information on the permitted shock and vibration resistance →  16, Section "Shock and vibration resistance".



A0003208

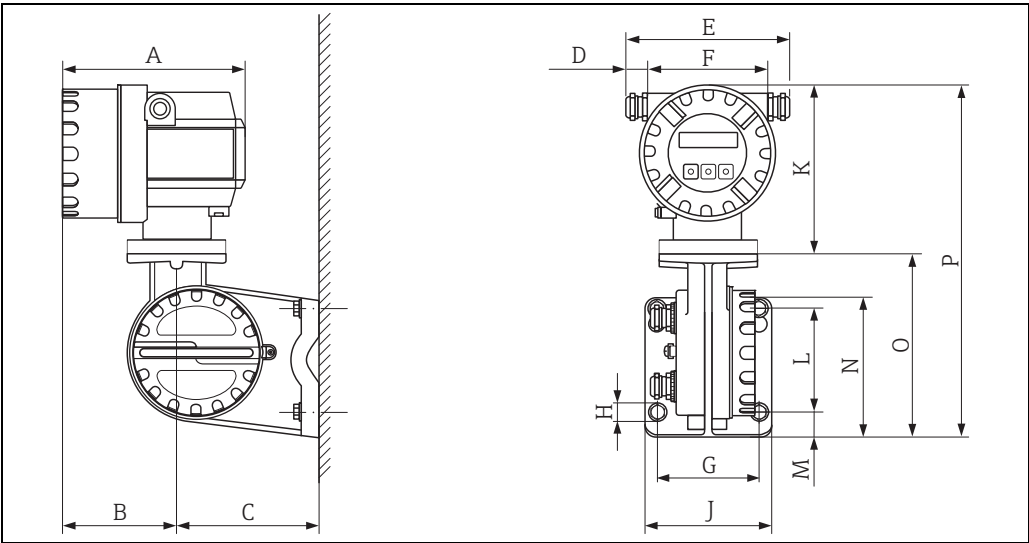
Measures to prevent vibration of the measuring device

$L > 10$ m (33 ft)

Mechanical construction

Design, dimensions

Transmitter, remote version



Transmitter dimensions, remote version

Dimensions in SI units

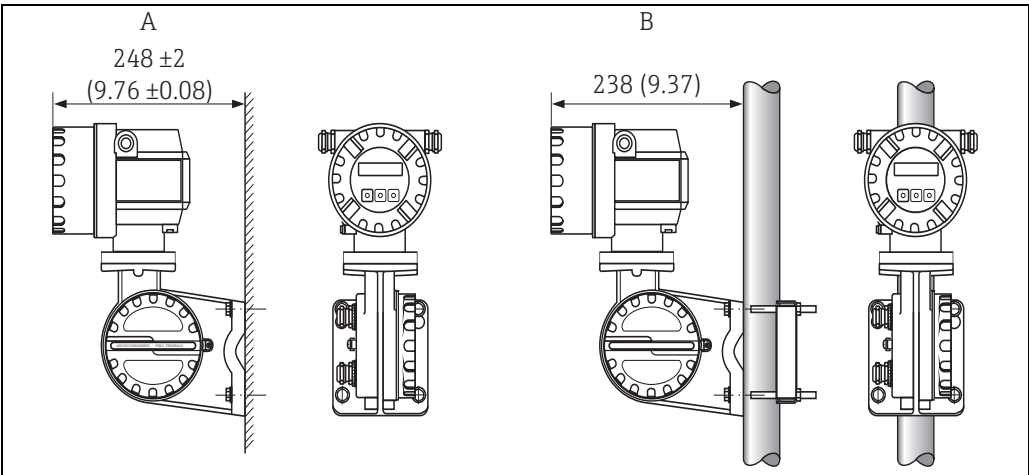
A	B	C	D	E	F	G	Ø H
178	113	135	20 to 30	161 to 181	113	100	8.6 (M8)
J	K	L	M	N	O	P	
123	150	100	25	133	177.5	327.5	

All dimensions in [mm]

Dimensions in US units

A	B	C	D	E	F	G	Ø H
7.00	4.45	5.31	0.79 to 1.81	6.34 to 7.13	4.44	3.94	0.34 (M8)
J	K	L	M	N	O	P	
4.84	5.90	3.94	0.98	5.24	6.99	12.89	

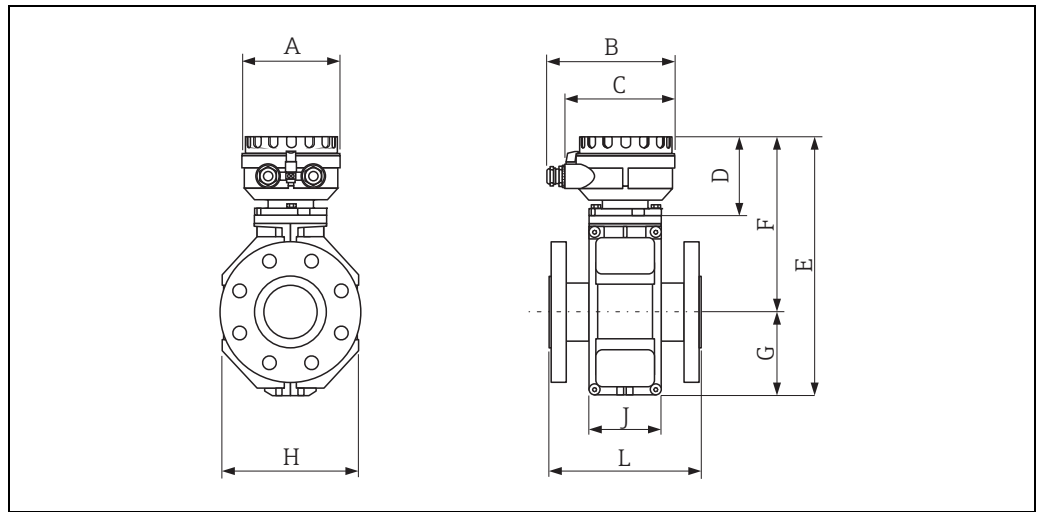
All dimensions in [inch]



Transmitter mounting, remote version. Engineering unit mm (in)

- A Direct wall mounting
- B Pipe mounting

Sensor, remote version, DN 25 to 300 (1 to 12")



A0012462

Dimensions in SI units

DN	L ¹⁾	A	B	C	D	E	F	G	H	J
25	200	129	163	143	102	286	202	84	120	94
32	200					286	202	84	120	94
40	200					286	202	84	120	94
50	200					286	202	84	120	94
65	200					336	227	109	180	94
80	200					336	227	109	180	94
100	250					336	227	109	180	94
125	250					417	267	150	260	140
150	300					417	267	150	260	140
200	350					472	292	180	324	156
250	450					522	317	205	400	156
300	500					572	342	230	460	166

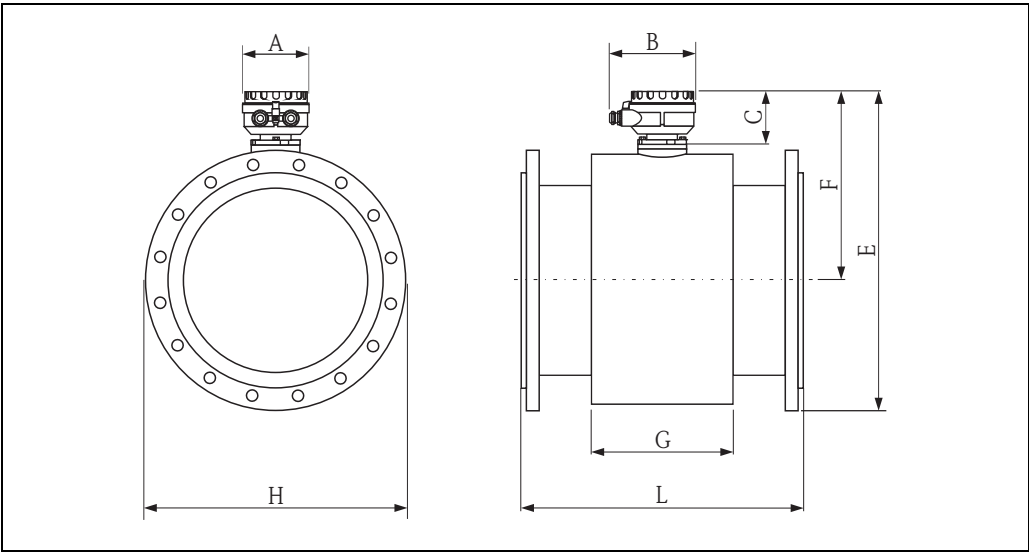
¹⁾ The length (L) is regardless of the pressure rating selected. Fitting length to DVGW.
All dimensions in [mm]

Dimensions in US units

DN	L ¹⁾	A	B	C	D	E	F	G	H	J
1"	7.87	5.08	6.42	5.63	4.02	11.3	7.95	3.32	4.72	3.70
1 ½"	7.87					11.3	7.95	3.32	4.72	3.70
2"	7.87					11.3	7.95	3.32	4.72	3.70
3"	7.87					13.2	8.94	4.30	7.10	3.70
4"	9.84					13.2	8.94	4.30	7.10	3.70
6"	11.8					16.4	10.5	5.91	10.2	5.51
8"	13.8					18.6	11.5	7.10	12.8	6.14
10"	17.7					20.6	12.5	8.08	15.8	6.14
12"	19.7					22.5	13.5	9.06	18.1	6.54

¹⁾ The length (L) is regardless of the pressure rating selected. Fitting length to DVGW.
All dimensions in [inch]

Sensor, remote version, DN 350 to 2400 (14 to 90")



A0014987

Dimensions in SI units

DN	L	A	B	C	F	G
350	550	129	163	102	353	290
375	600				379	290
400	600				379	290
450	600				407	290
500	600				432	290
600	600				473	290
700	700				538	424
750	750				575	454
800	800				594	500
900	900				644	580
1000	1000				694	664
1200	1200				808	832
1400	1400				920	1008
1600	1600				1020	1147
1800	1800				1128	1379
2000	2000				1239	1569
2200	2200				1339	1711
2400	2400				1444	1859

All dimensions in [mm]

DN	E with pressure rating					H with pressure rating				
	PN 6	PN 10	PN 16	ASME AWWA	AS	PN 6	PN 10	PN 16	ASME AWWA	AS
350	598	605	613	620	615	490	505	520	533	525
375	–	–	–	–	654	–	–	–	–	550
400	649	661	667	677	669	540	565	580	597	580
450	704	714	727	724	727	595	615	640	635	640
500	754	767	790	781	784	645	670	715	699	705
600	850	863	893	879	885	755	780	840	813	825
700	968	985	1003	1001	993	860	895	910	927	910
750	–	–	–	1067	1073	–	–	–	984	995
800	1082	1102	1107	1124	1124	975	1015	1025	1060	1060
900	1182	1202	1207	1228	1232	1075	1115	1125	1168	1175
1000	1282	1309	1322	1339	1322	1175	1230	1255	1289	1255
1200	1511	1536	1551	1564	1553	1405	1455	1255	1511	1490
1400	1735	1758	1763	–	–	1630	1675	1685	–	–
1600	1935	1978	1986	–	–	1830	1915	1930	–	–
1800	2150	2185	2193	2226	–	2045	2115	2130	2197	–
2000	2371	2401	2411	2420	–	2265	2325	2345	2362	–
2200	2576	2614	–	–	–	2475	2550	–	–	–
2400	2786	2824	–	–	–	2685	2760	–	–	–

All dimensions in [mm]

Dimensions in US units

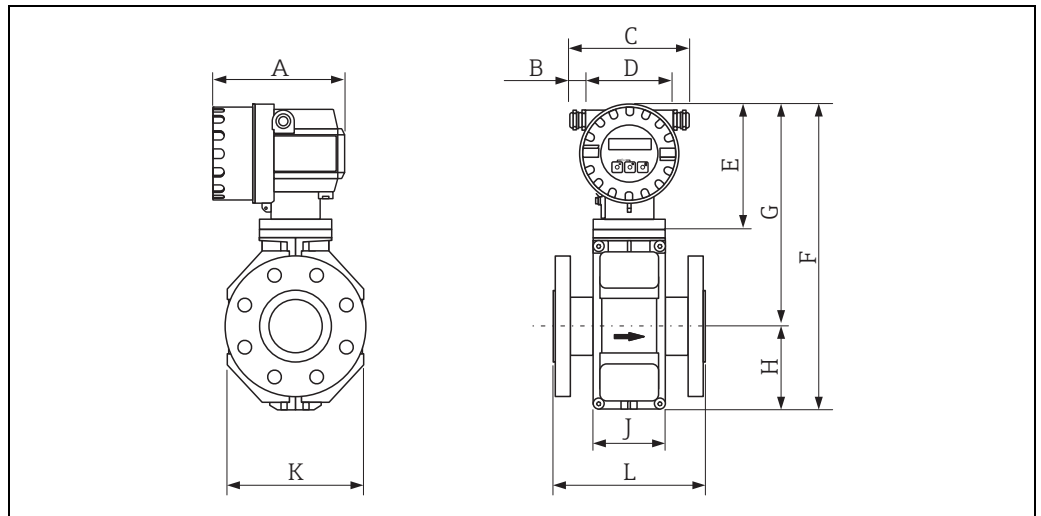
DN	L	A	B	C	F	G
14"	21.7	5.08	6.42	4.02	13.9	11.4
15"	23.6				14.9	11.4
16"	23.6				14.9	11.4
18"	23.6				16.0	11.4
20"	23.6				17.0	11.4
24"	23.6				18.6	11.4
28"	27.6				21.2	16.7
30"	29.5				22.6	17.9
32"	31.5				23.4	19.7
36"	35.4				25.4	22.8
40"	39.4				27.3	26.2
42"	41.3				28.7	29.9
48"	47.2				31.8	32.8
54"	53.2				36.2	39.7
60"	59.1				40.2	45.2
66"	65.0				42.2	50.6
72"	70.9				44.4	54.3
78"	78.7				48.8	61.8
84"	84.7				52.7	67.4
90"	90.6				56.8	73.2

All dimensions in [inch]

DN	E with pressure rating					H with pressure rating				
	PN 6	PN 10	PN 16	ASME AWWA	AS	PN 6	PN 10	PN 16	ASME AWWA	AS
14"	23.5	23.8	24.1	24.4	24.2	19.3	19.9	20.5	21.0	20.7
15"	–	–	–	–	25.7	–	–	–	–	21.7
16"	25.6	26.0	26.3	26.7	26.3	21.3	22.2	22.8	23.5	22.8
18"	27.7	28.1	28.6	28.5	28.6	23.4	24.2	25.2	25.0	25.2
20"	29.7	30.2	31.1	30.7	30.9	25.4	26.4	28.1	27.5	27.8
24"	33.5	34.0	35.2	34.6	34.8	29.7	30.7	33.1	32.0	32.5
28"	38.1	39.0	39.5	39.6	39.3	33.9	35.2	35.8	36.5	35.8
30"	–	–	–	42.2	42.4	–	–	–	38.7	39.2
32"	42.6	43.6	43.6	44.4	44.4	38.4	40.0	40.4	41.7	41.7
36"	46.5	47.5	47.5	48.5	48.7	42.3	43.9	44.3	46.0	46.3
40"	50.5	51.7	52.0	52.9	52.2	46.3	48.4	48.2	50.7	49.4
42"	–	–	–	55.4	–	–	–	–	53.0	–
48"	59.5	60.6	61.1	61.8	61.3	55.3	57.3	49.4	59.5	58.7
54"	–	–	69.4	69.4	–	–	–	66.3	66.3	–
60"	–	–	78.2	76.7	–	–	–	76.0	73.0	–
66"	–	–	–	82.2	–	–	–	–	80.0	–
72"	84.7	86.0	86.3	87.66	–	80.5	83.3	83.9	86.5	–
78"	93.4	94.5	94.9	95.3	–	89.2	91.5	92.3	93.0	–
84"	–	–	–	102.6	–	–	–	–	99.8	–
90"	–	–	–	110.1	–	–	–	–	106.5	–

All dimensions in [inch]

Compact version, DN 25 to 300 (1 to 12")



A0012464

Dimensions in SI units

DN	L ¹⁾	A	B	C	D	E	F	G	H	J	K
25	200	178	20 to 30	153 to 168	121	150	341	257	84	94	120
32	200						341	257	84	94	120
40	200						341	257	84	94	120
50	200						341	257	84	94	120
65	200						391	282	109	94	180
80	200						391	282	109	94	180
100	250						391	282	109	94	180
125	250						472	322	150	140	260
150	300						472	322	150	140	260
200	350						527	347	180	156	324
250	450	178	20 to 30	153 to 168	121	150	577	372	205	156	400
300	500						627	397	230	166	460

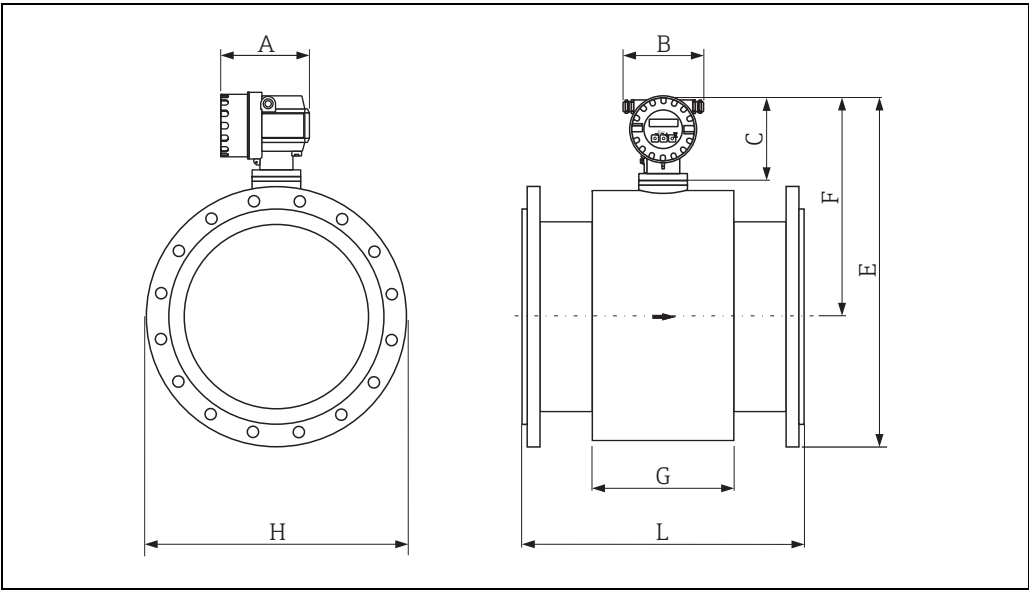
¹⁾ The length (L) is regardless of the pressure rating selected. Fitting length to DVGW.
All dimensions in [mm]

Dimensions in US units

DN	L ¹⁾	A	B	C	D	E	F	G	H	J	K
1"	7.87	7.01	0.79 to 1.81	6.02 to 6.61	4.76	5.91	13.4	10.1	3.32	3.70	4.72
1½"	7.87						13.4	10.1	3.32	3.70	4.72
2"	7.87						13.4	10.1	3.32	3.70	4.72
–	7.87						15.4	11.1	4.30	3.70	7.10
3"	7.87						15.4	11.1	4.30	3.70	7.10
4"	9.84						15.4	11.1	4.30	3.70	7.10
–	9.84						18.6	12.7	5.91	5.51	10.2
6"	11.8						18.6	12.7	5.91	5.51	10.2
8"	13.8						20.8	13.7	7.10	6.14	12.8
10"	17.7						22.7	14.7	8.08	6.14	15.8
12"	19.7	7.01	0.79 to 1.81	6.02 to 6.61	4.76	5.91	24.7	15.6	9.06	6.54	18.1

¹⁾ The length (L) is regardless of the pressure rating selected. Fitting length to DVGW.
All dimensions in [inch]

Compact version, DN 350 to 2400 (14 to 90")



A0014995

Dimensions in SI units

DN	L	A	B	C	F	G
350	550	178	161 to 181	150	401	290
375	600				427	290
400	600				427	290
450	600				455	290
500	600				480	290
600	600				521	290
700	700				591	424
750	750				628	454
800	800				647	500
900	900				697	580
1000	1000				747	664
1200	1200				861	832
1400	1400				920	1008
1600	1600				1020	1147
1800	1800				1128	1379
2000	2000				1239	1569
2200	2200				1339	1711
2400	2400				1444	1859

All dimensions in [mm]

DN	E at pressure rating					H at pressure rating				
	PN 6	PN 10	PN 16	ASME AWWA	AS	PN 6	PN 10	PN 16	ASME AWWA	AS
350	646	653	721	668	663	490	505	520	533	525
375	–	–	–	–	702	–	–	–	–	550
400	697	709	779	725	717	540	565	580	597	580
450	752	762	842	772	775	595	615	640	635	640
500	802	815	945	829	832	645	670	715	699	705
600	898	911	1045	927	933	755	780	840	813	825
700	1021	1038	1164	1054	1046	860	895	910	927	910
750	–	–	–	1120	1126	–	–	–	984	995
800	1135	1155	1259	1177	1177	975	1015	1025	1060	1060
900	1235	1255	1259	1281	1284	1075	1115	1125	1168	1175
1000	1335	1362	1374	1392	1374	1175	1230	1225	1289	1255
1200	1564	1588	1603	1617	1606	1405	1455	1255	1511	1490
1400	1735	1758	1815	–	–	1630	1675	1685	–	–
1600	1935	1978	2038	–	–	1830	1915	1930	–	–
1800	2150	2185	2245	2226	–	2045	2115	2130	2197	–
2000	2371	2401	2463	2420	–	2265	2325	2345	2362	–
2200	2576	2614	–	–	–	2475	2550	–	–	–
2400	2786	2824	–	–	–	2685	2760	–	–	–

All dimensions in [mm]

Dimensions in US units

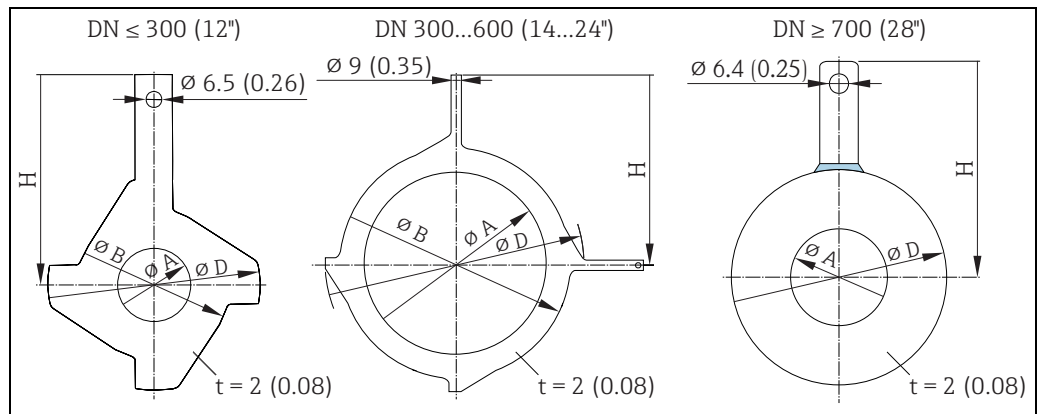
DN	L	A	B	C	F	G
14"	21.6	7.00	6.34 to 7.13	5.91	15.8	11.4
15"	23.6				16.8	11.4
16"	23.6				16.8	11.4
18"	23.6				17.9	11.4
20"	23.6				18.9	11.4
24"	23.6				20.5	11.4
28"	27.6				23.2	16.7
30"	29.5				24.7	17.9
32"	31.5				25.5	19.7
36"	35.4				27.4	22.8
40"	39.4				29.4	26.2
42"	41.3				30.8	29.9
48"	47.2				33.9	32.8
54"	53.1				36.2	39.7
60"	59.0				40.2	45.27
66"	64.9				42.2	50.6
72"	70.8				44.4	54.3
78"	78.7				48.8	61.8
84"	84.6				52.7	67.3
90"	90.5				56.9	73.2

All dimensions in [inch]

DN	E at pressure rating					H at pressure rating				
	PN 6	PN 10	PN 16	ASME AWWA	AS	PN 6	PN 10	PN 16	ASME AWWA	AS
14"	25.4	25.7	28.4	26.3	26.1	19.3	19.9	20.5	21.0	20.7
15"	–	–	–	–	27.6	–	–	–	–	21.7
16"	27.4	27.9	30.7	28.5	28.2	21.3	22.2	22.8	23.5	22.8
18"	29.8	30.0	33.1	30.4	30.5	23.4	24.2	25.2	25.0	25.2
20"	31.6	32.0	37.2	32.6	32.8	25.4	26.4	28.1	27.5	27.8
24"	35.4	35.9	41.1	36.5	36.7	29.7	30.7	33.1	32.0	32.5
28"	40.2	40.93	45.8	41.5	41.2	33.9	35.2	35.8	36.5	35.8
30"	–	–	–	44.1	44.3	–	–	–	38.7	39.2
32"	44.7	45.5	49.6	46.3	46.3	38.4	40.0	40.4	41.7	41.7
36"	48.6	49.4	49.6	50.4	50.6	42.3	43.9	44.3	46.0	46.3
40"	52.5	53.6	54.1	54.8	54.1	46.3	48.4	48.2	50.7	49.4
42"	–	–	–	57.3	–	–	–	–	53.0	–
48"	61.6	62.5	63.1	63.6	63.2	55.3	57.3	49.4	59.5	58.7
54"	–	–	71.5	71.3	–	–	–	66.3	66.3	–
60"	–	–	80.2	78.5	–	–	–	76.0	72.9	–
66"	–	–	–	84.0	–	–	–	–	80.0	–
72"	86.5	87.9	88.4	89.5	–	80.5	83.2	83.9	86.5	–
78"	95.2	96.4	97.0	97.2	–	89.1	91.5	92.3	92.9	–
84"	–	–	–	104.5	–	–	–	–	–	99.7
90"	–	–	–	111.9	–	–	–	–	–	106.5

All dimensions in [inch]

Ground disk



A0015442

Engineering unit mm (in)

Dimensions in SI and US units

DN		Pressure rating	A		B		D		H	
[mm]	[inch]		[mm]	[inch]	[mm]	[inch]	[mm]	[inch]	[mm]	[inch]
25	1"	1)	26	1.02	62	2.44	77.5	3.05	87.5	3.44
32	–	1)	35	–	80	–	87.5	–	94.5	–
40	1 ½"	1)	41	1.61	82	3.23	101.0	3.98	103	4.06
50	2"	1)	52	2.05	101	3.98	115.5	4.55	108	4.25
65	–	1)	68	2.68	121	4.76	131.5	5.18	118	4.65
80	3"	1)	80	3.15	131	5.16	154.5	6.08	135	5.31
100	4"	1)	104	4.09	156	6.14	186.5	7.34	153	6.02
125	–	1)	130	5.12	187	7.36	206.5	8.13	160	6.30
150	6"	1)	158	6.22	217	8.54	256	10.08	184	7.24
200	8"	1)	206	8.11	267	10.51	288	11.34	205	8.07
250	10"	1)	260	10.24	328	12.91	359	14.13	240	9.45
300	12"	1)	312	12.28	375	14.76	413	16.26	273	10.75
350	14"	DIN, PN 6	343	13.50	420	16.54	479	18.86	365	14.37
		DIN, PN 10/16								
		ASME, Cl.150								
375	15"	PN 16	393	15.5	461	18.15	523	20.6	395	15.6
400	16"	DIN, PN 6	393	15.47	470	18.50	542	21.34	395	15.55
		DIN, PN 10/16								
		ASME, Cl.150								
450	18"	DIN, PN 6	439	17.28	525	20.67	583	22.95	417	16.42
		DIN, PN 10/16								
		ASME, Cl.150								
500	20"	DIN, PN 6	493	19.41	575	22.64	650	25.59	460	18.11
		DIN, PN 10/16								
		ASME, Cl.150								
600	24"	DIN, PN 6	593	23.35	676	26.61	766	30.16	522	20.55
		DIN, PN 10/16								
		ASME, Cl.150								

DN		Pressure rating	A		B		D		H	
[mm]	[inch]		[mm]	[inch]	[mm]	[inch]	[mm]	[inch]	[mm]	[inch]
700	28"	DIN, PN 6	697	27.44	–	–	786	30.94	460	18.11
		DIN, PN 10	693	27.28	–	–	813	32.01	480	18.9
		AS, PN 16	687	27.05	–	–	807	31.77	490	19.29
		AWWA, Class D	693	27.28	–	–	832	32.76	494	19.45
750	30"	AS, PN 16								
		AWWA, Class D	743	29.25	–	–	833	32.8	523	20.59
800	32"	DIN, PN 6	799	31.46	–	–	893	35.16	520	20.47
		DIN, PN 10	795	31.30	–	–	920	36.22	540	21.26
		AS, PN 16	789	31.06	–	–	914	35.98	550	21.65
		AWWA, Class D	795	31.30	–	–	940	37.01	561	22.09
900	36"	DIN, PN 6	897	35.31	–	–	993	39.09	570	22.44
		DIN, PN 10	893	35.16	–	–	1020	40.16	590	23.23
		AS, PN 16	886	34.88	–	–	1014	39.92	595	23.43
		AWWA, Class D	893	35.16	–	–	1048	41.26	615	24.21
1000	40"	DIN, PN 6	999	39.33	–	–	1093	43.03	620	24.41
		DIN, PN 10	995	39.17	–	–	1127	44.37	650	25.59
		AS, PN 16	988	38.90	–	–	1131	44.53	660	25.98
		AWWA, Class D	995	39.17	–	–	1163	45.79	675	26.57
–	42"	AWWA, Class D	1044	41.10	–	–	1220	48.03	704	27.72
1200	48"	DIN, PN 6	1203	47.36	–	–	1310	51.57	733	28.86
		DIN, PN 10	1196	47.09	–	–	1344	52.91	760	29.92
		AS, PN 16	1196	47.09	–	–	1385	54.53	786	30.94
		AWWA, Class D	1188	46.77	–	–	1345	52.95	775	30.51

1) Ground disks can be used for all flange norms/ pressure ratings.

Weight

SI units

Compact version (lap joint flanges / welded flanges DN > 300/12")

Weight data in kg		Compact version (including transmitter)											
Nominal diameter		EN (DIN)						ASME/ AWWA		AS			
[mm]	[inch]												
25	1"	PN 6	-	PN 10	-	PN 16	7.3	ASME / Class 150	7.9	PN 16	-	Table E	-
32	-		-		-		8.0		-		-		
40	1 ½"		-		-		9.0		7.5		-		
50	2"		-		-		9.4		7.6		-		
65	-		-		-		10.4		-		-		
80	3"		-		-		12.4		12.8		-		
100	4"		-		-		14.4		16.1		-		
125	-		-		-		15.9		-		-		
150	6"		-		-		23.9		24.4		-		
200	8"		-		43.4		44.9		49.6		-		
250	10"		-		63.4		70.7		75.1		-		
300	12"		-		68.4		85.8		100		-		
350	14"		77.4		88.4		106		137		99.4		99.4
375	15"		-		-		-		-		105		-
400	16"		89.4		104		124		168		124		120
450	18"		104		119		150		191		142		152
500	20"		114		132		191		228		191		182
600	24"		155		182		301		327		283		281
700	28"		215		274		335	278	386		350		
750	30"		-		-		-	338	470		458		
800	32"		289		374		462	402	569		518		
900	36"		384		476		582	498	739		739		
1000	40"		493		615		795	666	854		856		
-	42"		-		-		-	771	-		-		
1200	48"		707		916		1314	1035	1368		1368		
-	54"		-		-		-	1438	-		-		
1400	-		1126		1482		1906	-	-		-		
-	60"		-		-		-	1785	-		-		
1600	-		1521		2197		2698	-	-		-		
-	66"		-		-		-	2463	-		-		
1800	72"		2001		2838		3687	2857	-		-		
-	78"		2777		3508		4646	3532	-		-		
2000	-		2777		3508		4646	3532	-		-		
-	84"		-		-		-	3883	-		-		
2200	-		3065		4172		-	-	-		-		
-	90"		-		-		-	4847	-		-		
2400	-		3940		5035		-	-	-		-		
Transmitter Promag (compact version): 1.8 kg (Weight data valid without packaging material)													

Remote version (lap joint flanges / welded flanges DN > 300/12")

Weight data in kg													
Nominal diameter		Remote version (sensor plus sensor housing without cable)											
[mm]	[inch]	EN (DIN)						ASME/ AWWA		AS			
25	1"	PN 6	–	PN 10	–	PN 16	5.3	ASME / Class 150	5.9	PN 16	–	Table E	–
32	–		–		–		6.0		–		–		
40	1 ½"		–		–		7.0		5.5		–		
50	2"		–		–		7.4		5.6		–		
65	–		–		–		8.4		–		–		
80	3"		–		–		10.4		10.8		–		
100	4"		–		–		12.4		14.1		–		
125	–		–		–		13.9		–		–		
150	6"		–		–		21.9		22.4		–		
200	8"		–		41.4		42.9		47.6		–		
250	10"		–		61.4		68.7		73.1		–		
300	12"		–		66.4		83.8		98		–		
350	14"		75.4		86.4		103		139		97.4		
375	15"		–		102		–		–		103		
400	16"		87.4		102		121		170		123		
450	18"		103		118		149		193		141		
500	20"		112		130		190		230		190		
600	24"		156		181		300		329		282		
700	28"		214		273		334	278	385				
750	30"		–		–		–	339	471				
800	32"		288		373		461	402	568				
900	36"		383		475		581	498	738				
1000	40"		492		614		794	666	853				
–	42"		–		–		–	771	–				
1200	48"		706		915		1313	1035	1367				
–	54"		–		–		–	1438	–				
1400	–		1125		1381		1905	–	–				
–	60"		–		–		–	1785	–				
1600	–		1520		2196		2697	–	–				
–	66"		–		–		–	2463	–				
1800	72"		2000		2837		3686	2857	–				
–	78		2776		2837		4645	3532	–				
2000	–		2776		3507		4645	3532	–				
–	84"		–		–		–	3883	–				
2200	–		3064		4171		–	–	–				
–	90"		–		–		–	4847	–				
2400	–		3939		5034		–	–	–				
Transmitter Promag (remote version): 3.1 kg (Weight data valid without packaging material)													

Lap joint flanges, stamped plate

Weight data in kg						
Nominal diameter		Compact version		Remote version (without cable)		
[mm]	[inch]			Sensor EN (DIN)		Transmitter
25	1"	PN 10	5.8	PN 10	3.8	4.2
32	–		5.4		3.4	4.2
40	1 ½"		6.3		4.7	4.2
50	2"		5.4		3.4	4.2
65	–		6.2		4.2	4.2
80	3"		7.2		5.2	4.2
100	4"		9.7		7.7	4.2
125	–		13.2		11.2	4.2
150	6"		17.2		15.2	4.2
200	8"		35.7		33.7	4.2
250	10"		54.2		52.2	4.2
300	12"		55.2		53.2	4.2
Transmitter Promag (compact version): 1.8 kg						
(Weight data valid for standard pressure ratings and without packaging material)						

US units

ASME/AWWA (lap joint flanges / welded flanges DN > 300/12")

Weight data in lbs					
Nominal diameter		Compact version		Remote version	
[mm]	[inch]	ASME/AWWA		ASME/AWWA	
25	1"	ASME / Class 150	17.4	ASME / Class 150	13
32	–		–		–
40	1 ½"		16.5		12.1
50	2"		16.8		12.3
65	–		–		–
80	3"		28.2		23.8
100	4"		35.5		31.1
125	–		–		–
150	6"		53.8		49.4
200	8"		109		105
250	10"		166		161
300	12"		221		216
350	14"		302		306
375	15"		–		–
400	16"		370		274
450	18"		421		425
500	20"		503		507
600	24"	726	725		
700	28"	AWWA / Class D	613	AWWA / Class D	612
750	30"		745		746
800	32"		886		885
900	36"		1098		1097
1000	40"		1468		1467
–	42"		1701		1700
1200	48"		2283		2282
–	54"		3171		3170
1400	–		–		–
–	60"		3935		3934
1600	–		–		–
–	66"		5430		5429
1800	72"		6300		6299
–	78"		7787		7786
2000	–		7787		–
–	84"		8561		8560
2200	–		–		–
–	90"	10686	10685		
2400	–	–	–		
Transmitter Promag (compact version): 4.0 lbs Transmitter Promag (remote version): 6.8 lbs (Weight data valid without packaging material)					

Measuring tube specifications

Internal diameter for pressure rating EN (DIN), AS 2129, AS 4087, ASME and AWWA

Nominal diameter		Pressure rating			Internal diameter measuring tube					
		EN (DIN)	AS 2129 AS 4087	ASME AWWA	Hard rubber		Polyurethane		PTFE	
[mm]	[inch]				[mm]	[inch]	[mm]	[inch]	[mm]	[inch]
25	1"	PN 10/16	–	Class 150	–	–	23.7	0.95	25.3	1.0
32	–	PN 10/16	–	Class 150	–	–	32.4	1.28	34.0	1.3
40	1 ½"	PN 10/16	–	Class 150	–	–	38.3	1.5	39.9	1.6
50	2"	PN 10/16	–	Class 150	–	–	50.3	2.0	51.9	2.0
65*	–	PN 10/16	–	Class 150	–	–	66.1	2.6	67.7	2.7
80	3"	PN 10/16	–	Class 150	–	–	78.9	3.1	79.9	3.1
100	4"	PN 10/16	–	Class 150	–	–	104.3	4.1	103.8	4.1
125	–	PN 10/16	–	Class 150	–	–	129.7	5.1	129.1	5.1
150	6"	PN 10/16	–	Class 150	–	–	158.3	6.2	156.3	6.2
200	8"	PN 10	–	–	–	–	206.7	8.1	202.1	8.0
		PN 16	–	Class 150	–	–	205.1	8.1	200.5	7.9
250	10"	PN 10	–	–	–	–	260.6	10.3	256.2	10.1
		PN 16	–	Class 150	–	–	259.0	10.2	254.6	10.0
300	12"	PN 10	–	–	–	–	311.5	12.3	305.5	12.0
		PN 16	–	Class 150	–	–	309.9	12.2	303.9	12.0
350	14"	PN 6	–	–	341	13.4	344	13.5	–	–
		PN 10	–	–	341	13.4	344	13.5	–	–
		PN 16	–	–	337	13.2	340	13.3	–	–
		–	PN 16 Table E	–	337	13.2	340	13.3	–	–
		–	–	Class 150	339	13.3	342	13.4	–	–
375	15"	–	PN 16	–	389	15.3	392	15.4	–	–
400	16"	PN 6	–	–	391	15.4	394	15.5	–	–
		PN 10	–	–	391	15.4	394	15.5	–	–
		PN 16	–	–	389	15.3	392	15.4	–	–
		–	PN 16 Table E	–	389	15.3	392	15.4	–	–
		–	–	Class 150	387	15.2	390	15.4	–	–
450	18"	PN 6	–	–	442	17.4	445	17.5	–	–
		PN 10	–	–	442	17.4	445	17.5	–	–
		PN 16	–	–	440	17.3	443	17.4	–	–
		–	PN 16 Table E	–	440	17.3	443	17.4	–	–
		–	–	Class 150	436	17.1	439	17.2	–	–
500	20"	PN 6	–	–	493	19.4	496	19.5	–	–
		PN 10	–	–	493	19.4	494	19.3	–	–
		PN 16	–	–	489	19.2	492	19.3	–	–
		–	PN 16 Table E	–	489	19.2	492	19.3	–	–
		–	–	Class 150	487	19.1	490	19.2	–	–
600	24"	PN 6	–	–	595	23.4	598	23.5	–	–
		PN 10	–	–	593	23.3	596	23.4	–	–
		PN 16	–	–	591	23.3	594	23.4	–	–
		–	PN 16 Table E	–	591	23.3	594	23.4	–	–
		–	–	Class 150	585	23.0	588	23.1	–	–
700	28"	PN 6	–	–	696	27.4	699	27.5	–	–
		PN 10	–	–	694	27.3	697	27.4	–	–
		PN 16	–	–	694	27.2	693	27.3	–	–
		–	PN 16 Table E	–	690	27.2	693	27.3	–	–
		–	–	Class D	694	27.3	697	27.4	–	–
750	30"	–	PN 16 Table E	–	741	29.2	744	29.3	–	–
		–	–	Class D	743	29.3	746	29.4	–	–

Nominal diameter		Pressure rating			Internal diameter measuring tube					
		EN (DIN)	AS 2129 AS 4087	ASME AWWA	Hard rubber		Polyurethane		PTFE	
[mm]	[inch]				[mm]	[inch]	[mm]	[inch]	[mm]	[inch]
800	32"	PN 6	–	–	796	31.3	799	31.5	–	–
		PN 10	–	–	794	31.2	797	31.4	–	–
		PN 16	–	–	788	31.0	791	31.1	–	–
		–	PN 16 Table E	–	788	31.0	791	31.1	–	–
		–	–	Class D	794	31.3	797	31.4	–	–
900	36"	PN 6	–	–	895	35.2	898	35.4	–	–
		PN 10	–	–	893	35.1	896	35.2	–	–
		PN 16	–	–	889	35.0	892	35.1	–	–
		–	PN 16 Table E	–	889	35.0	892	35.1	–	–
		–	–	Class D	895	35.2	898	35.4	–	–
1000	40"	PN 6	–	–	997	39.2	1000	39.3	–	–
		PN 10	–	–	995	39.1	998	39.3	–	–
		PN 16	–	–	991	39.0	994	39.1	–	–
		–	PN 16 Table E	–	991	39.0	994	39.1	–	–
		–	–	Class D	995	39.2	998	39.3	–	–
–	42"	–	–	Class D	1046	41.2	1049	41.3	–	–
1200	48"	PN 6	–	–	1201	47.3	1204	47.4	–	–
		PN 10	–	–	1199	47.2	1202	47.3	–	–
		PN 16	–	–	1191	46.9	1194	47.0	–	–
		–	PN 16 Table E	–	1191	46.9	1194	47.0	–	–
		–	–	Class D	1195	47.0	1198	47.2	–	–
–	54"	–	–	Class D	1346	53.0	–	–	–	–
1400	–	PN 6	–	–	1401	55.1	–	–	–	–
		PN 10	–	–	1395	54.9	–	–	–	–
		PN 16	–	–	1387	54.6	–	–	–	–
–	60"	–	–	Class D	1499	59.0	–	–	–	–
1600	–	PN 6	–	–	1599	62.9	–	–	–	–
		PN 10	–	–	1591	62.6	–	–	–	–
		PN 16	–	–	1581	62.2	–	–	–	–
–	66"	–	–	Class D	1647	53.0	1650	64.9	–	–
1800	72"	PN 6	–	–	1799	70.8	1802	70.9	–	–
		PN 10	–	–	1791	70.5	1794	70.6	–	–
		PN 16	–	–	1781	70.1	–	–	–	–
		–	–	Class D	1791	70.5	1794	70.6	–	–
–	78"	PN 6	–	–	1999	78.7	–	–	–	–
		PN 10	–	–	1991	78.4	–	–	–	–
		PN 16	–	–	1979	77.9	–	–	–	–
		–	–	Class D	1987	78.2	–	–	–	–
2000	–	PN 6	–	–	1995	78.5	–	–	–	–
		PN 10	–	–	1991	78.4	–	–	–	–
		PN 16	–	–	1979	77.9	–	–	–	–
		–	–	Class D	1987	78.2	–	–	–	–
–	84"	–	–	Class D	2100	82.7	–	–	–	–
2200	–	PN 6	–	–	2195	86.4	–	–	–	–
		PN 10	–	–	2187	86.1	–	–	–	–
		–	–	Class D	1987	78.2	–	–	–	–
–	90"	–	–	Class D	2247	88.5	–	–	–	–
2400	–	PN 6	–	–	2391	94.1	–	–	–	–
		PN 10	–	–	2387	94.0	–	–	–	–

* Designed acc. to EN 1092-1 (not to DIN 2501)

Material

- Transmitter housing:
 - Compact housing: powder-coated die-cast aluminum
 - Wall-mounted housing: powder-coated die-cast aluminum
- Sensor housing
 - DN 25 to 300 (1 to 12"): powder-coated die-cast aluminum
 - DN 350 to 1200 (14 to 48"): with protective lacquering
- Measuring tube:
 - DN 25 to 300 (1 to 12"): Stainless-steel, 1.4301/1.4306 (304L)
 - DN 350 to 1200 (14 to 48"): Stainless-steel, 1.4301/1.4307 (304)
 - DN 1350 to 2400 (54 to 90"): Stainless-steel, 1.4301/1.4307
- **Electrodes: 1.4435 (316L); Alloy C22, 2.4602 (UNS N06022)**
- Flanges (All lap joint flanges made from carbon steel are hot-dip galvanized):
 - EN 1092-1 (DIN 2501)
 - DN 25 to 300
 - Lap joint flange:
 - Stainless-steel, 1.4306/1.4307
 - Carbon steel, 235JR
 - Lap joint flange, stampel plate:
 - Stainless-steel, 1.4301 (304)
 - Carbon steel, RSt37-2
 - DN 350 to 2400: carbon steel, S235JRG2, S235JR+N, P250GH, P245GH, E250C, A105
 - DN 350 to 600: Stainless-steel, 1.4571
 - DN 700 to 1000: Stainless-steel, 1.4404
 - ASME B16.5
 - DN ≤ 300 (12"), lap joint flange:
 - Stainless-steel, F316L
 - Carbon steel, A105
 - DN ≥ 350 (14"):
 - Carbon steel, A105
 - Stainless-steel, F316L
 - AWWA C207: A105, A181 Cl.70, E250C, S235JRG2, P265GH, S275JR
 - AS 2129: Carbon steel, A105, P235GH, P265GH, S235JRG2, E250C
 - AS 4087: Carbon steel, A105, P265GH, S275JR, E250C
- Seals: to DIN EN 1514-1 IBC form
- Ground disks: 1.4435 (316L) or Alloy C22, 2.4602 (UNS N06022)

Fitted electrodes

Measuring electrodes, reference electrodes and empty pipe detection electrodes available as standard with:

- 1.4435 (316L)
- Alloy C22, 2.4602 (UNS N06022)

Process connections

- EN 1092-1 (DIN 2501)
 - (Dimensions to DIN 2501, DN 65 PN 16 exclusively to EN 1092-1)
 - DN ≤ 300: lap joint flange (PN 10/16), lap joint flange, stamped plate (PN 10) = form A
 - DN ≥ 350: fixed flange (PN 6/10)= flat race
- ASME B16.5
 - DN ≤ 300 (12"): lap joint flange(Class 150)
 - DN ≥ 350 (14"): fixed flange (Class 150)
- AWWA C207
 - DN 48 to 90": fixed flange (Class D)
- AS 2129
 - DN 350 to 1200: fixed flange (Table E)
- AS 4087
 - DN 350 to 1200: fixed flange (PN16)

For information on the materials of the process connections →  37 Material

Surface roughness

Electrodes with 1.4435 (316L), Alloy C22, 2.4602 (UNS N06022): ≤ 0.3 to 0.5 µm (11.8 to 19.7 µin)
(all data refer to parts in contact with medium)

Operability

Local operation

Display elements

- Liquid crystal display: unilluminated, two lines with 16 characters per line
- Display (operating mode) preconfigured: volume flow and totalizer status
- 1 totalizers

Operating elements

- Onsite operation via three keys (◻, ⊕, ⊞)

Remote operation

Operation via HART protocol and FieldCare

Certificates and approvals

CE mark

The measuring system is in conformity with the statutory requirements of the EC Directives. Endress+Hauser confirms successful testing of the device by affixing to it the CE mark.

C-tick mark

The measuring system meets the EMC requirements of the "Australian Communication and Media Authority (ACMA)".

Drinking water approval

- WRAS BS 6920
- ACS
- NSF 61
- KTW/W270

Other standards and guidelines

- EN 60529
Degrees of protection by housing (IP code).
- EN 61010
Protection Measures for Electrical Equipment for Measurement, Control, Regulation and Laboratory Procedures.
- IEC/EN 61326
"Emission in accordance with requirements for Class A".
Electromagnetic compatibility (EMC requirements).
- ANSI/ISA-S82.01
Safety Standard for Electrical and Electronic Test, Measuring, Controlling and related Equipment - General Requirements. Pollution degree 2, Installation Category II.
- CAN/CSA-C22.2 No. 1010.1-92
Safety requirements for Electrical Equipment for Measurement and Control and Laboratory Use. Pollution degree 2, Installation Category II

Ordering information

Detailed ordering information is available from the following sources:

- In the Product Configurator on the Endress+Hauser website: www.endress.com → Select country → Instruments → Select device → Product page function: Configure this product
- From your Endress+Hauser Sales Center: www.endress.com/worldwide



Note!

Product Configurator - the tool for individual product configuration

- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Accessories

Various accessories, which can be ordered separately from Endress+Hauser, are available for the transmitter and the sensor. Your Endress+Hauser service organization can provide detailed information on the order codes in question.

Device-specific accessories

For the transmitter

Accessory	Description
Promag 10 transmitter	Transmitter for replacement or for stock. Use the order code to define the following specifications: ■ Approvals ■ Degree of protection / version ■ Cable type for the remote version ■ Cable entries ■ Display / power supply / operation ■ Software ■ Outputs / inputs
Mounting set	Mounting set for the aluminum field housing (remote version). Suitable for pipe mounting.
Cable for remote version	Coil current and electrode cables, various lengths.

For the sensor

Accessory	Description
Ground cable	Ground cable for potential equalization.
Ground disk	Ground disk for potential equalization.

Communication-specific accessories

Accessory	Description
HART handheld terminal Field Xpert SFX 100	Handheld terminal for remote configuration and for obtaining measured values via the HART current output (4 to 20 mA). Contact your Endress+Hauser representative for more information.
Fieldgate FXA320	Gateway for remote interrogation of HART sensors and actuators via web browser: ■ 2-channel analog input (4 to 20 mA) ■ 4 binary inputs with event counter function and frequency measurement ■ Communication via modem, Ethernet or GSM ■ Visualization via Internet/Intranet in the web browser and/or WAP cellular phone ■ Limit value monitoring with alarm signaling by e-mail or SMS ■ Synchronized time stamping of all measured values.
Fieldgate FXA520	Gateway for remote interrogation of HART sensors and actuators via web browser: ■ Web server for remote monitoring of up to 30 measuring points ■ Intrinsically safe version [EEx ia]IIC for applications in hazardous areas ■ Communication via modem, Ethernet or GSM ■ Visualization via Internet/Intranet in the web browser and/or WAP cellular phone ■ Limit value monitoring with alarm signaling by e-mail or SMS ■ Synchronized time stamping of all measured values ■ Remote diagnostics and remote configuration of connected HART devices
FXA195	The Commubox FXA195 connects intrinsically safe smart transmitters using the HART protocol to the USB port of a personal computer. This enables remote operation of the transmitter with operating software (e.g. FieldCare). Power is supplied to the Commubox via the USB port.

Service-specific accessories

Accessory	Description
Applicator	Software for selecting and configuring flowmeters. Applicator can be downloaded from the Internet or ordered on CD-ROM for installation on a local PC. Contact your Endress+Hauser representative for more information.
Fieldcheck	Tester/simulator for testing flowmeters in the field. When used in conjunction with the "FieldCare" software package, test results can be imported into a database, printed and used for official certification. Contact your Endress+Hauser representative for more information.
FieldCare	FieldCare is Endress+Hauser's FDT-based plant asset management tool. It can configure all smart field units in your system and helps you manage them. By using the status information, it is also a simple but effective way of checking their status and condition.
FXA193	The FXA193 service interface connects the device to the PC for operation via FieldCare.
Memograph M graphic display recorder	The Memograph M graphic display recorder provides information on all the relevant process variables: Measured values are recorded correctly, limit values are monitored and measuring points analyzed. The data are stored in the 256 MB internal memory and also on a SD card or USB stick. The ReadWin® 2000 PC software is part of the standard package and is used for configuring, visualizing and archiving the data captured.

Documentation

- Flow measurement (FA00005D/06)
- Operating Instructions Promag 10 (BA00082D/06)

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